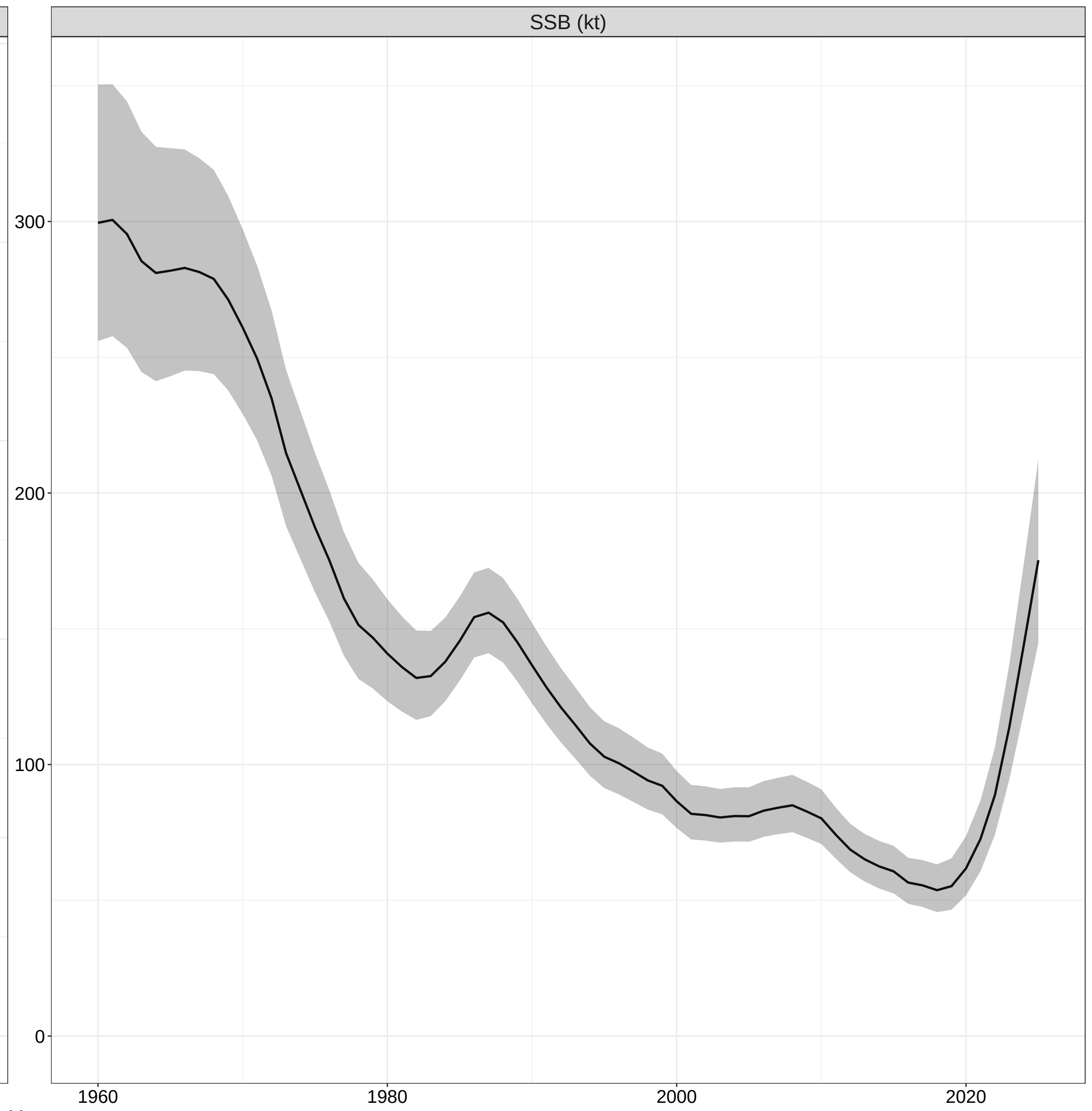
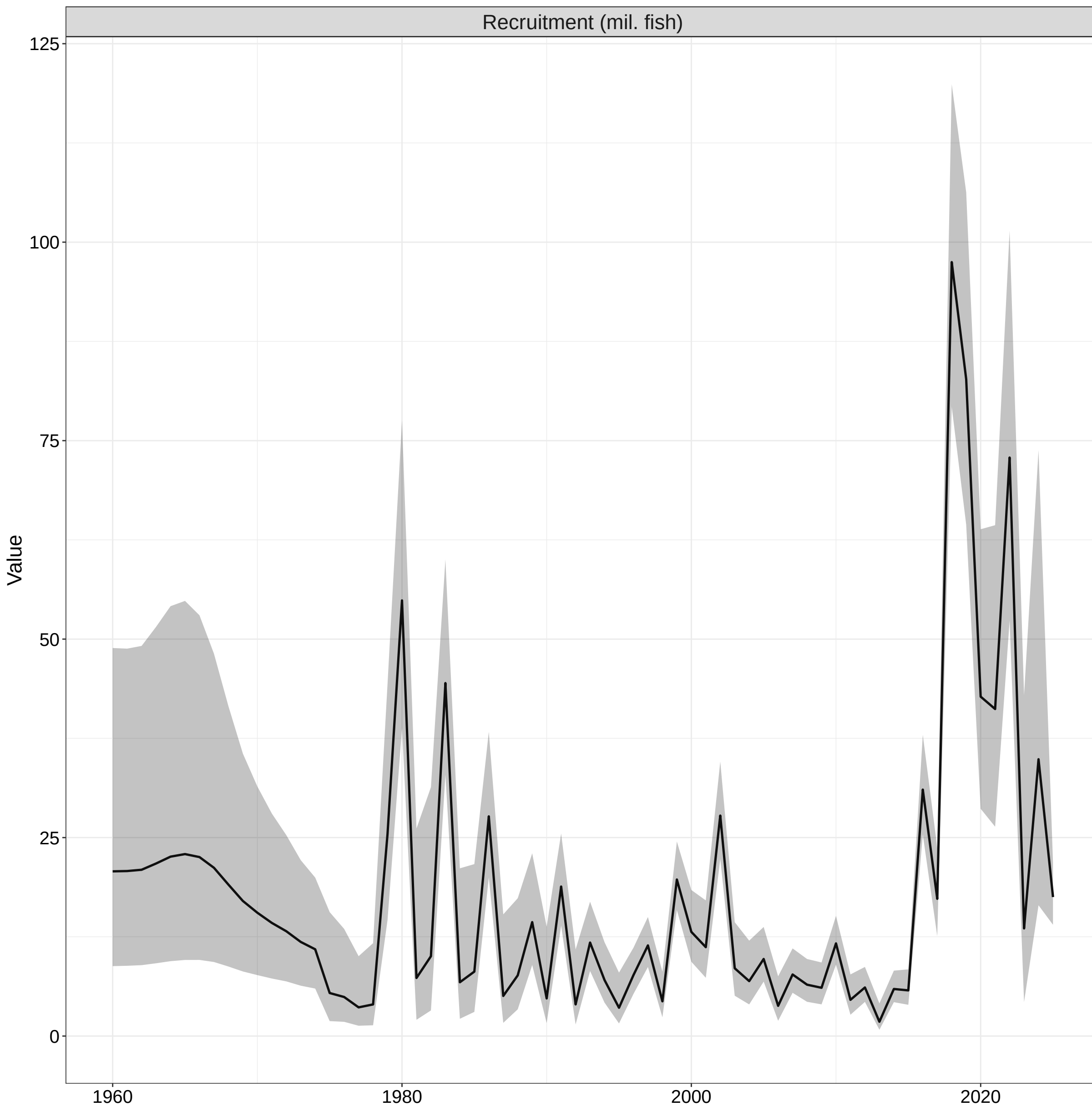
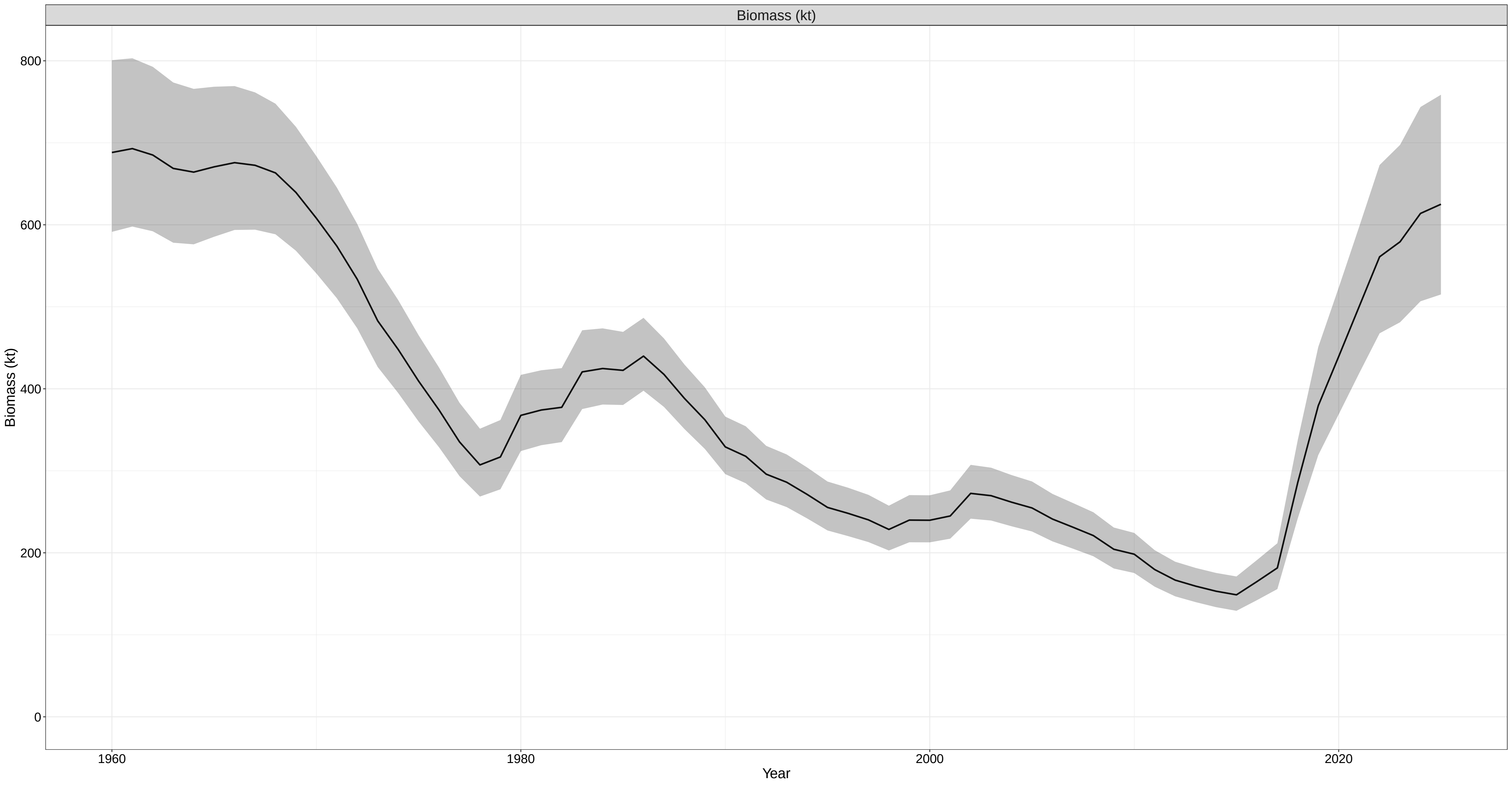


|   | Model | Region | Terminal_SSB | Terminal_SSB0 | Terminal_F        | Catch_Advice | B_Ref_Pt  | F_Ref_Pt | B_over_B_Ref | B_over_DynB_Ref | F_over_F_Ref |
|---|-------|--------|--------------|---------------|-------------------|--------------|-----------|----------|--------------|-----------------|--------------|
| 1 | 1     | 1      | 175.25956    | 340.55845     | 0.056173568707296 | 32.42408     | 133.85402 | 0.0753   | 1.30933      | 0.51462         | 0.74602      |



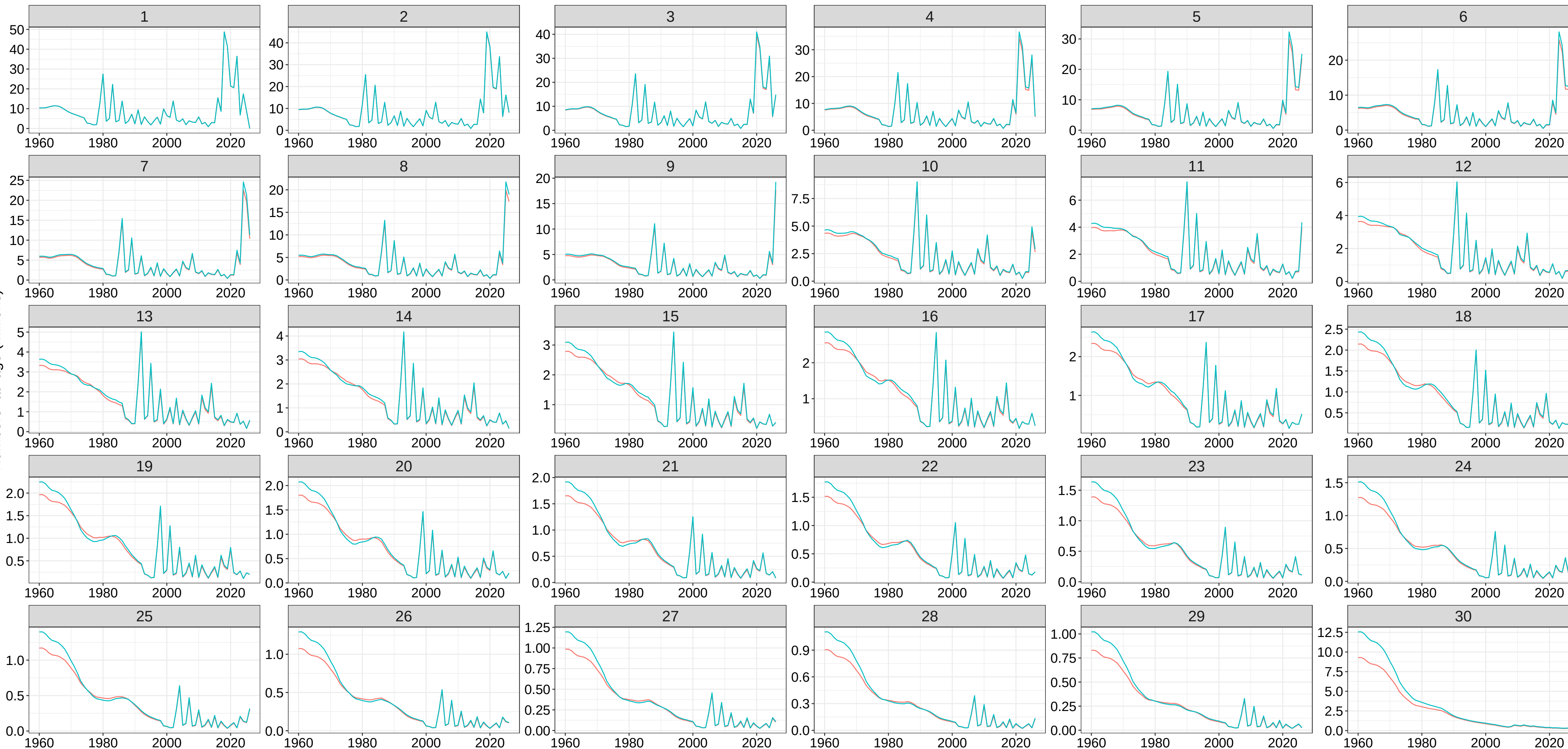




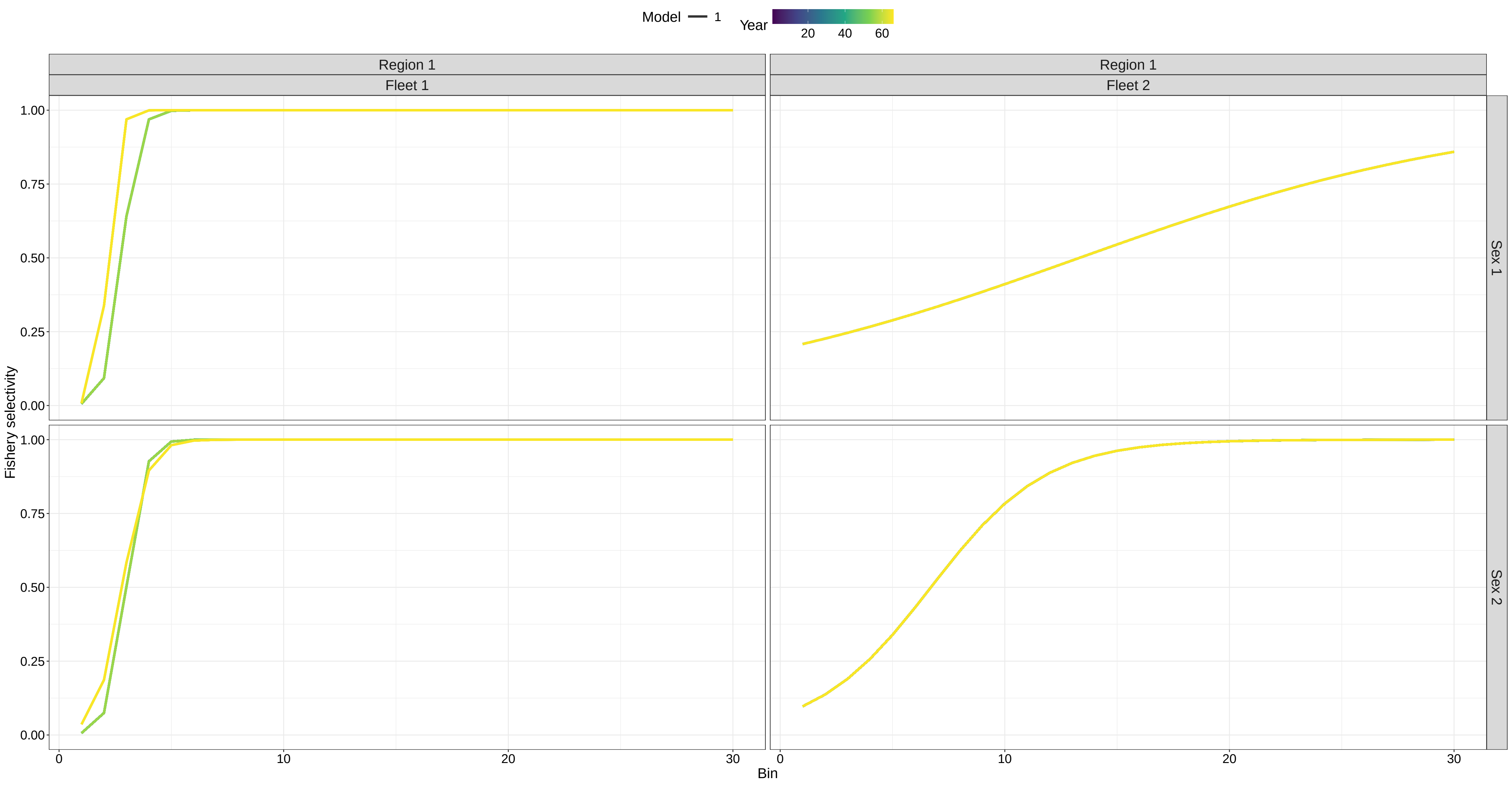
# Numbers-at-Age

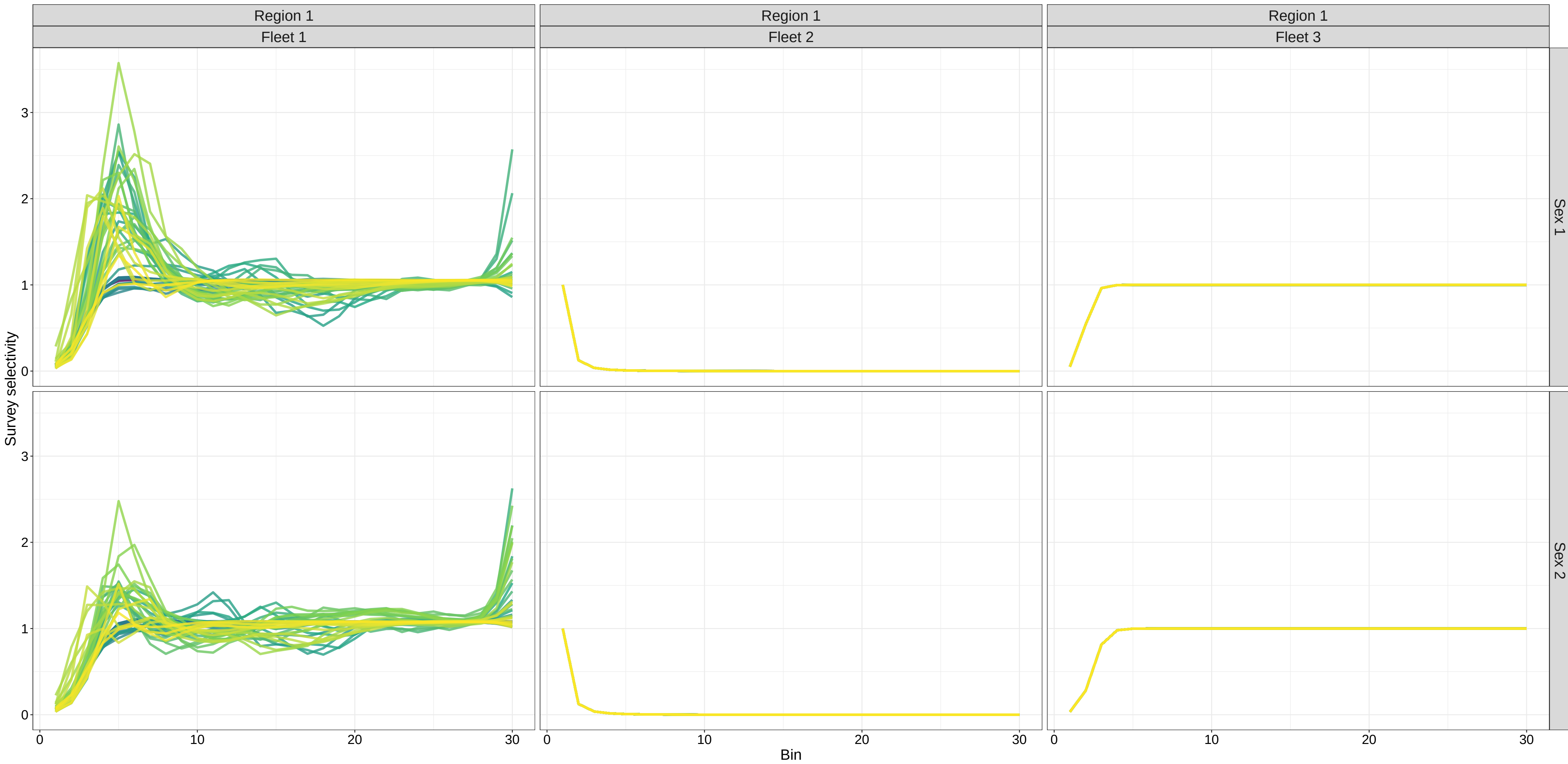
Sex — Female — Male

Numbers-at-age (millions)

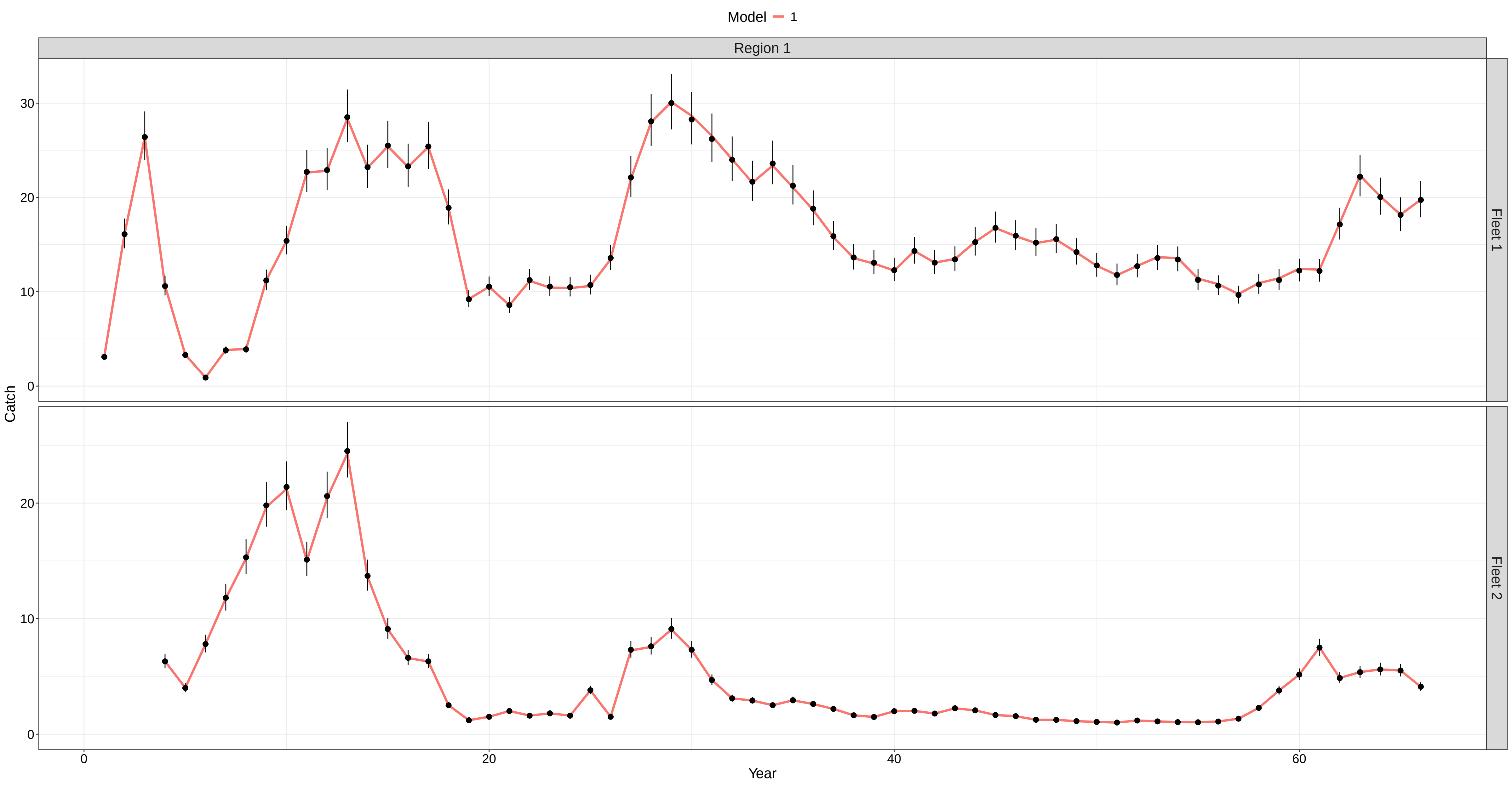


Year

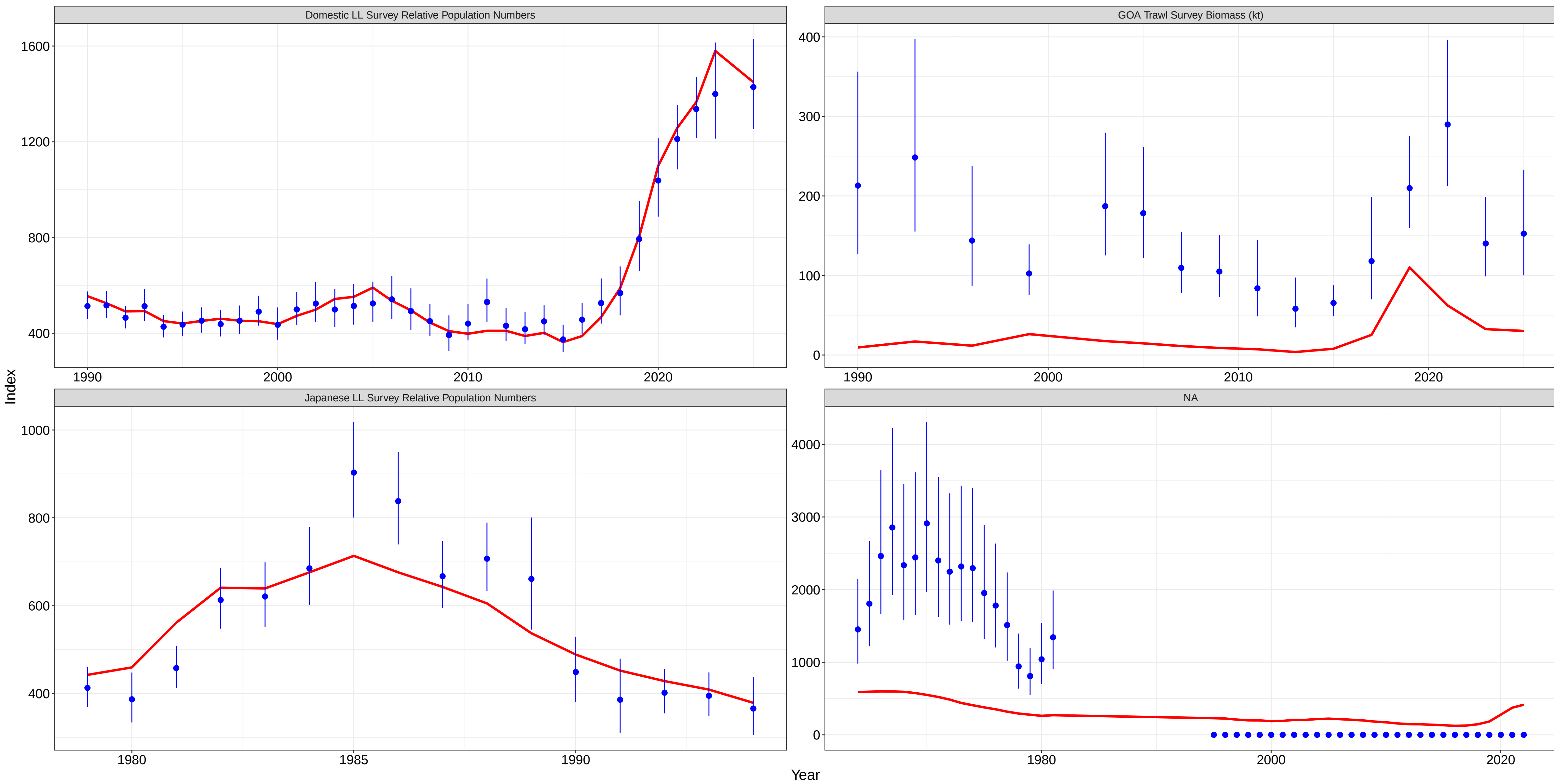






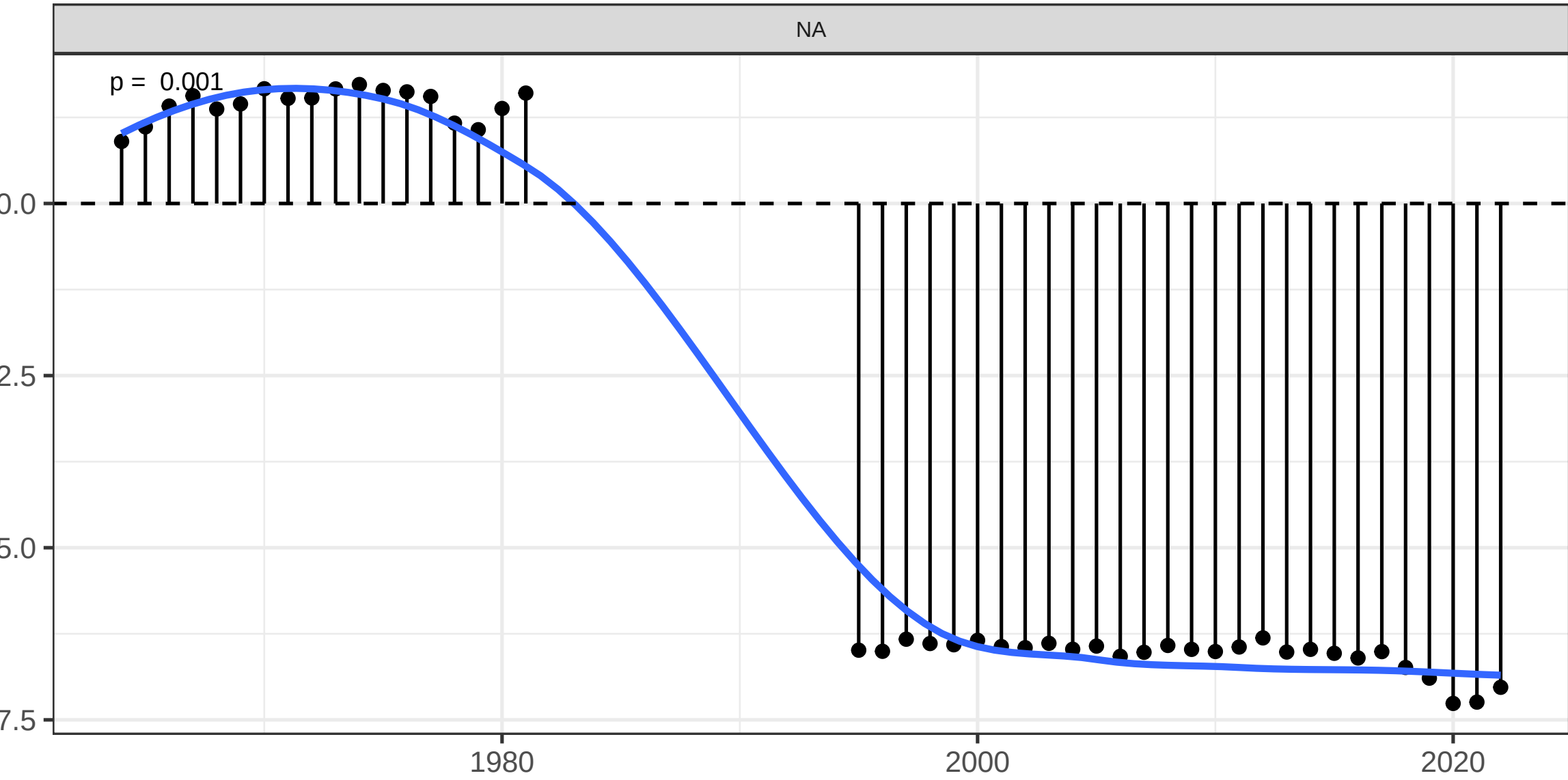
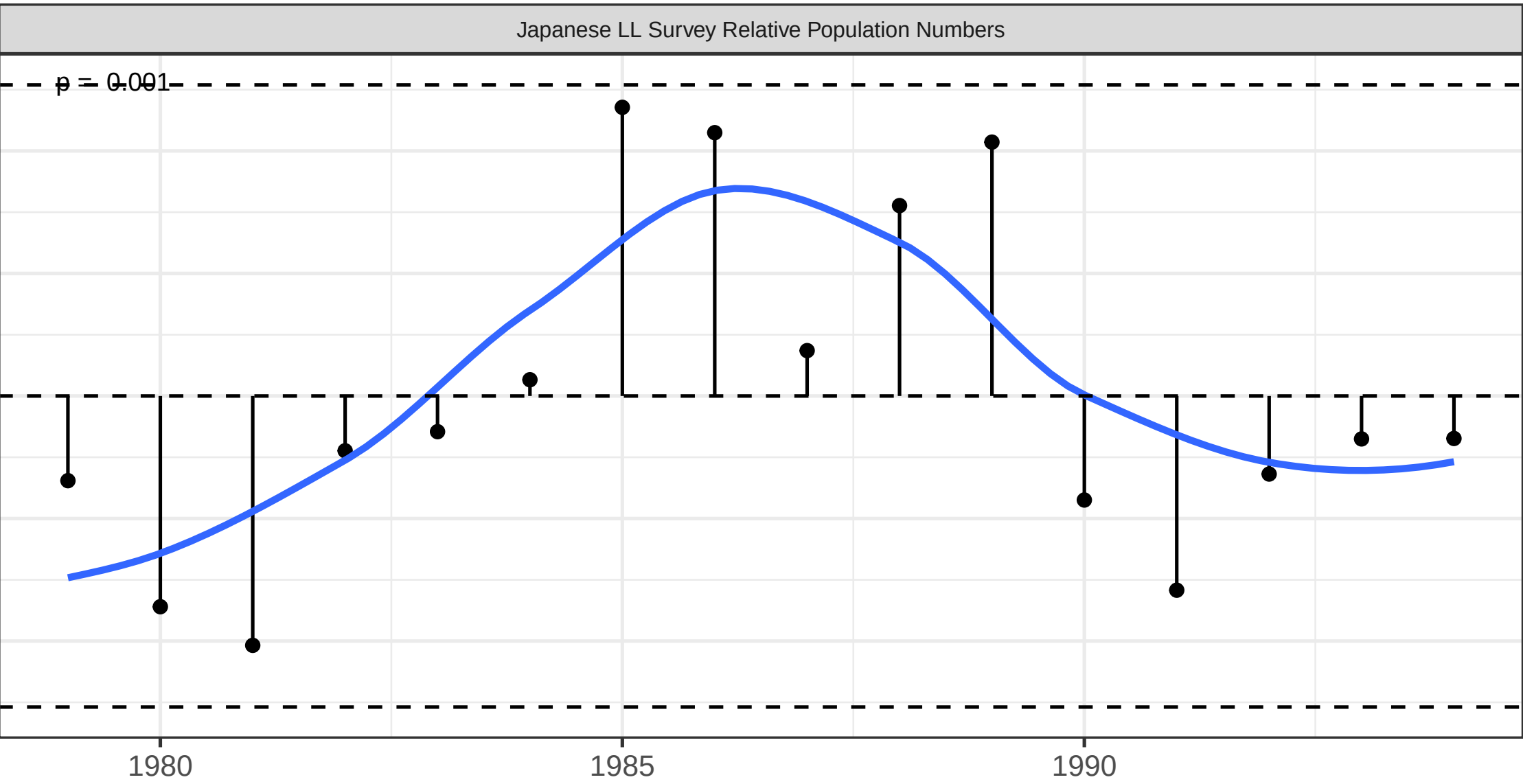
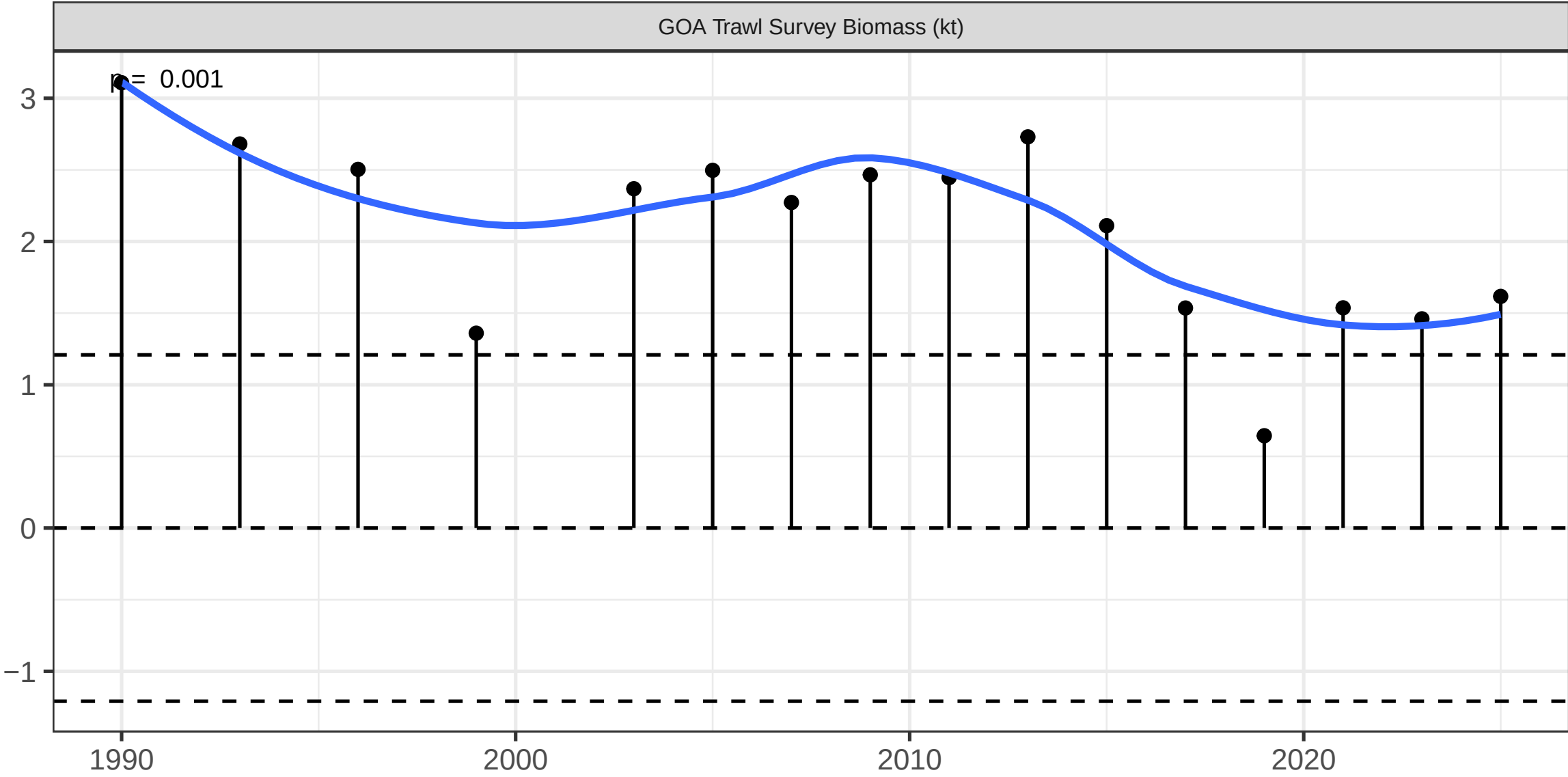
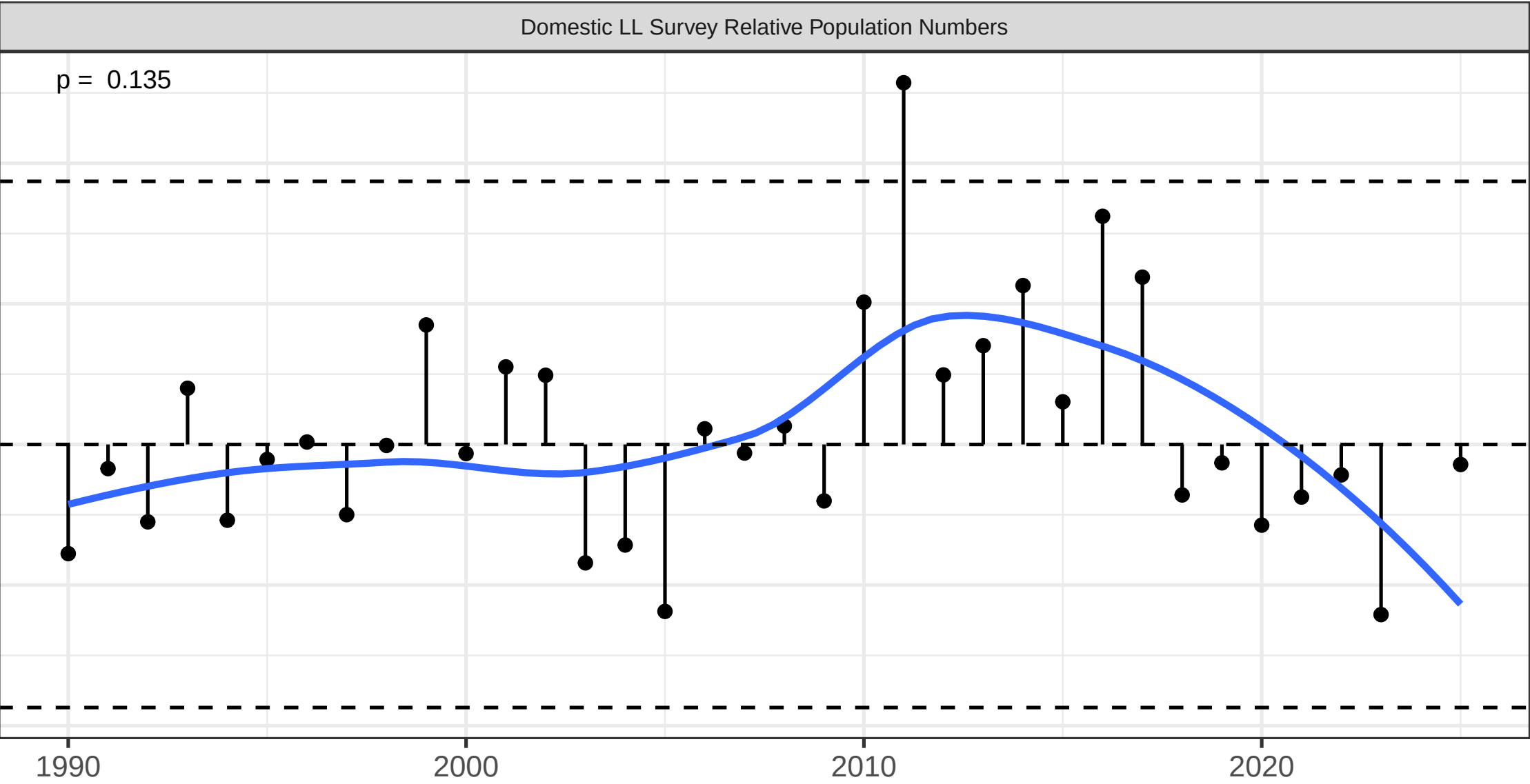


# Index Fits



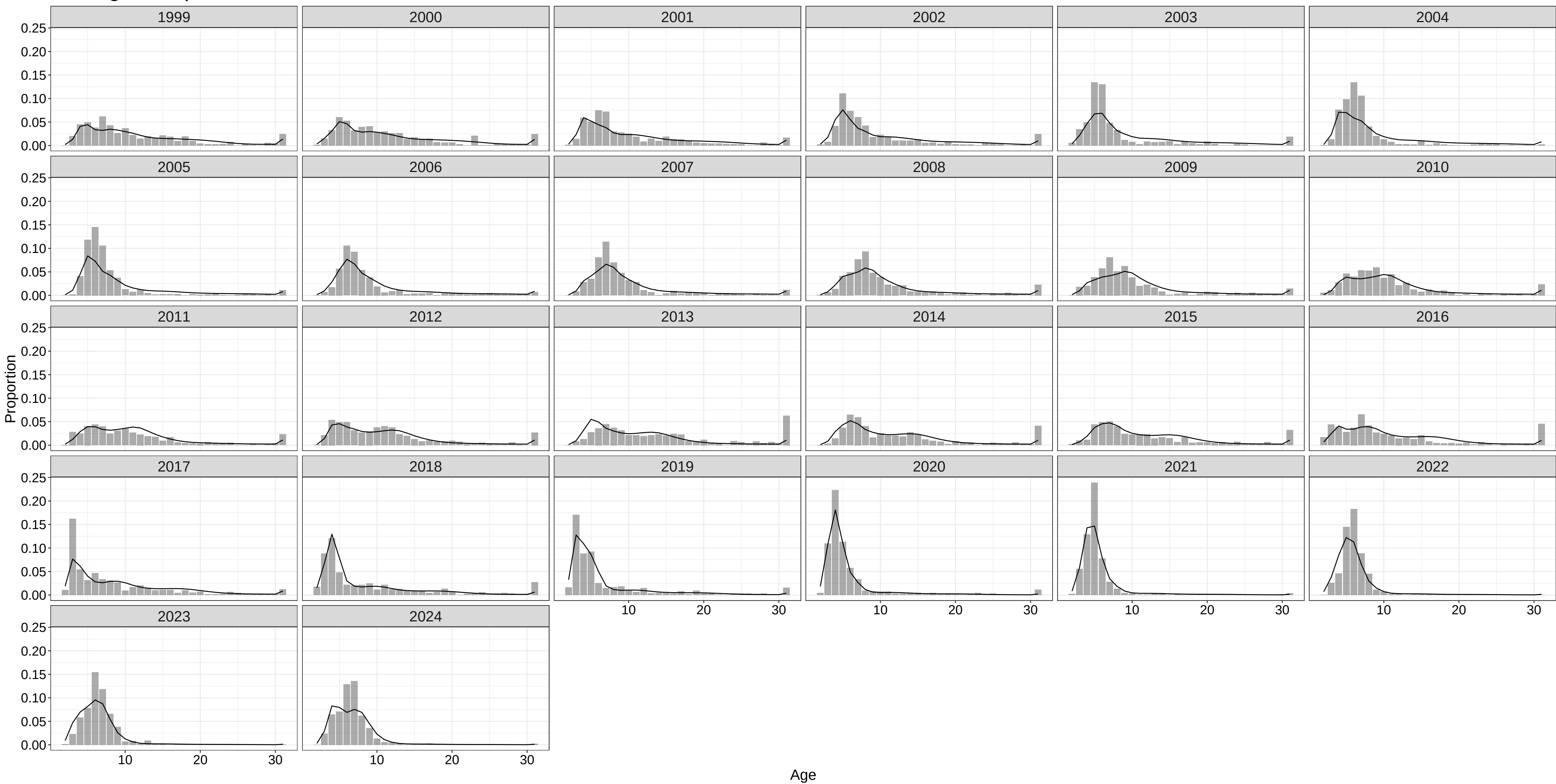


# Index Residuals Runs Test

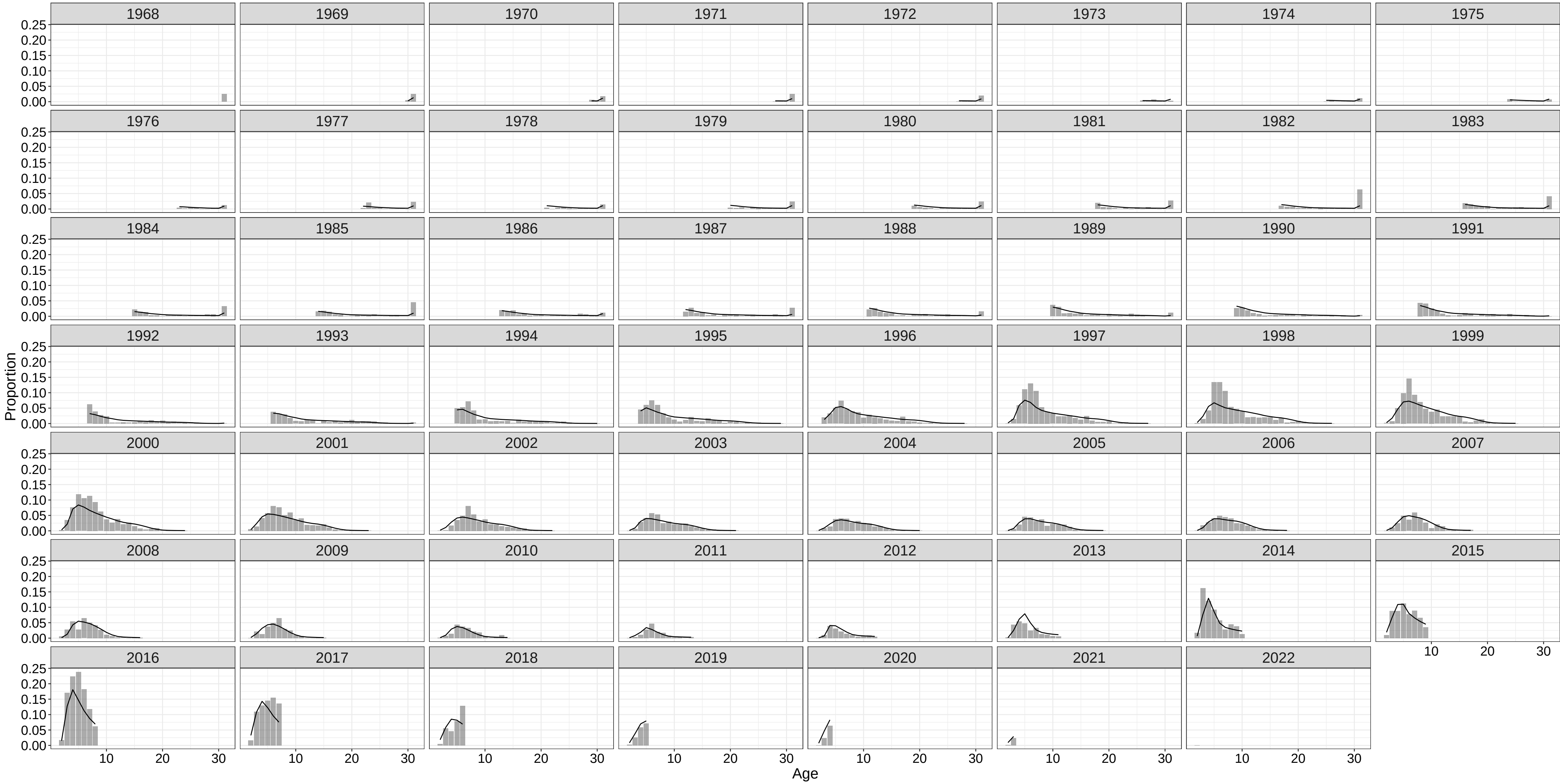


Year

# LLF Age Comps Female

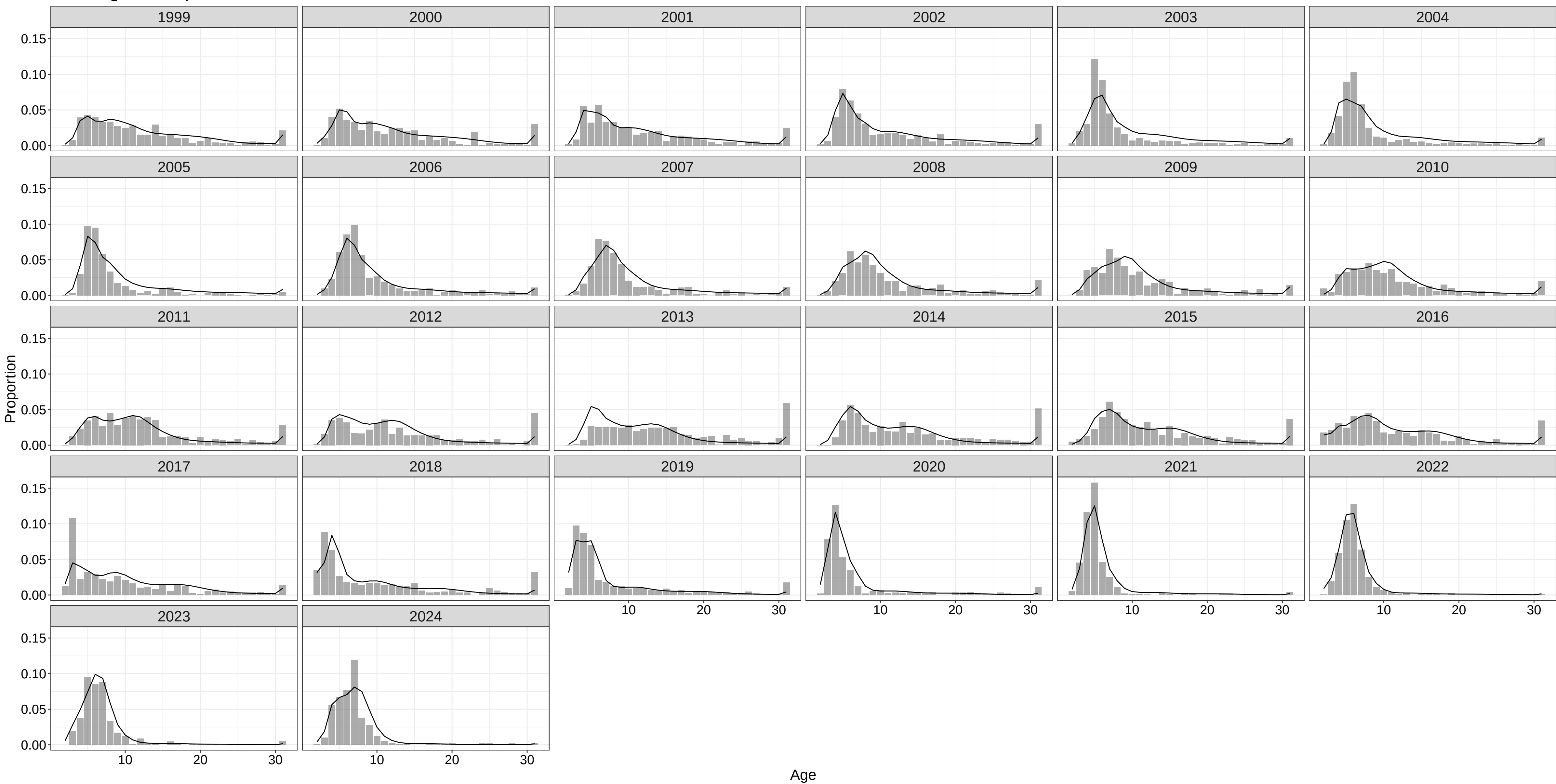


LLF Age Comps Cohort Fits Female

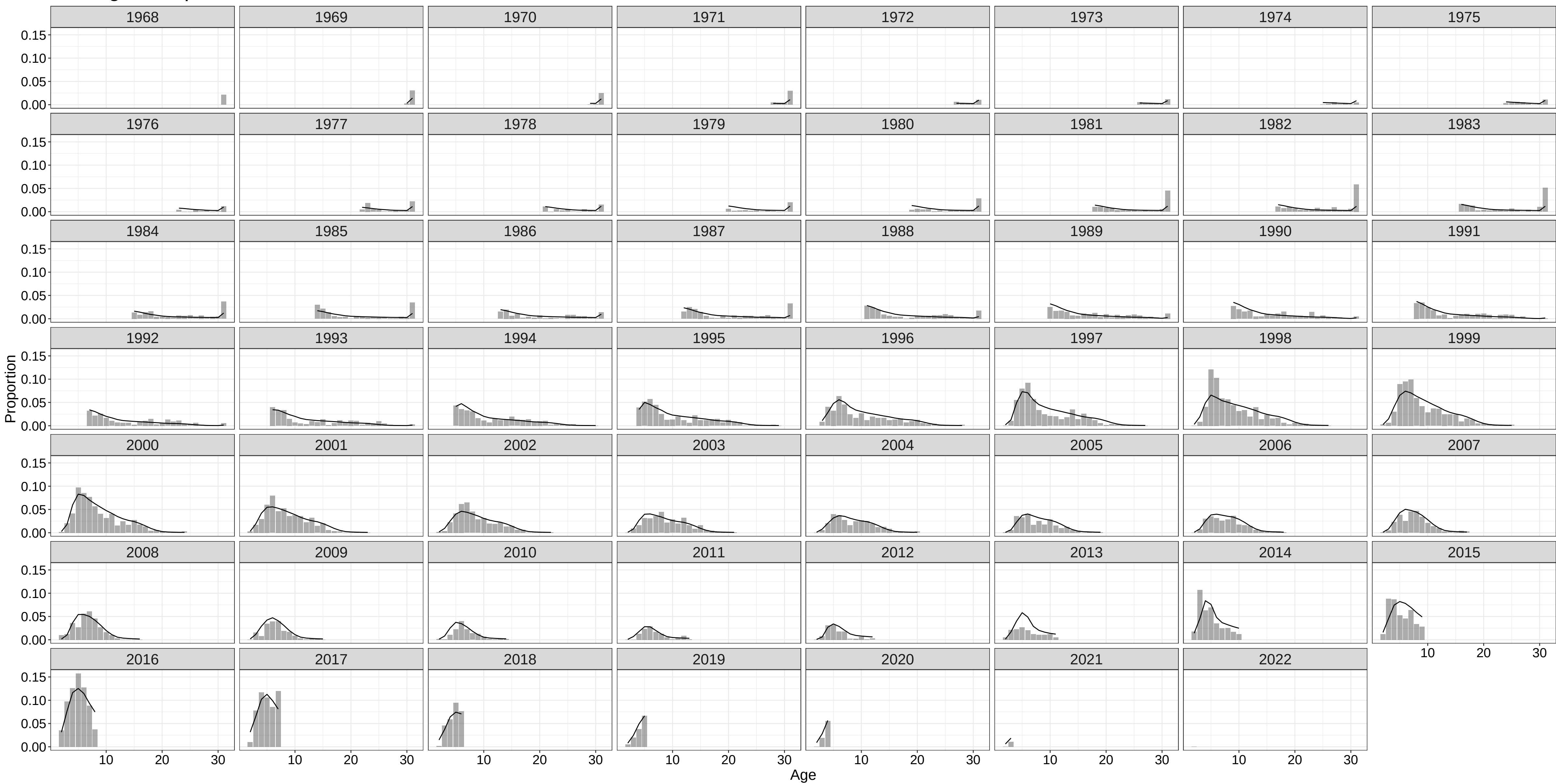




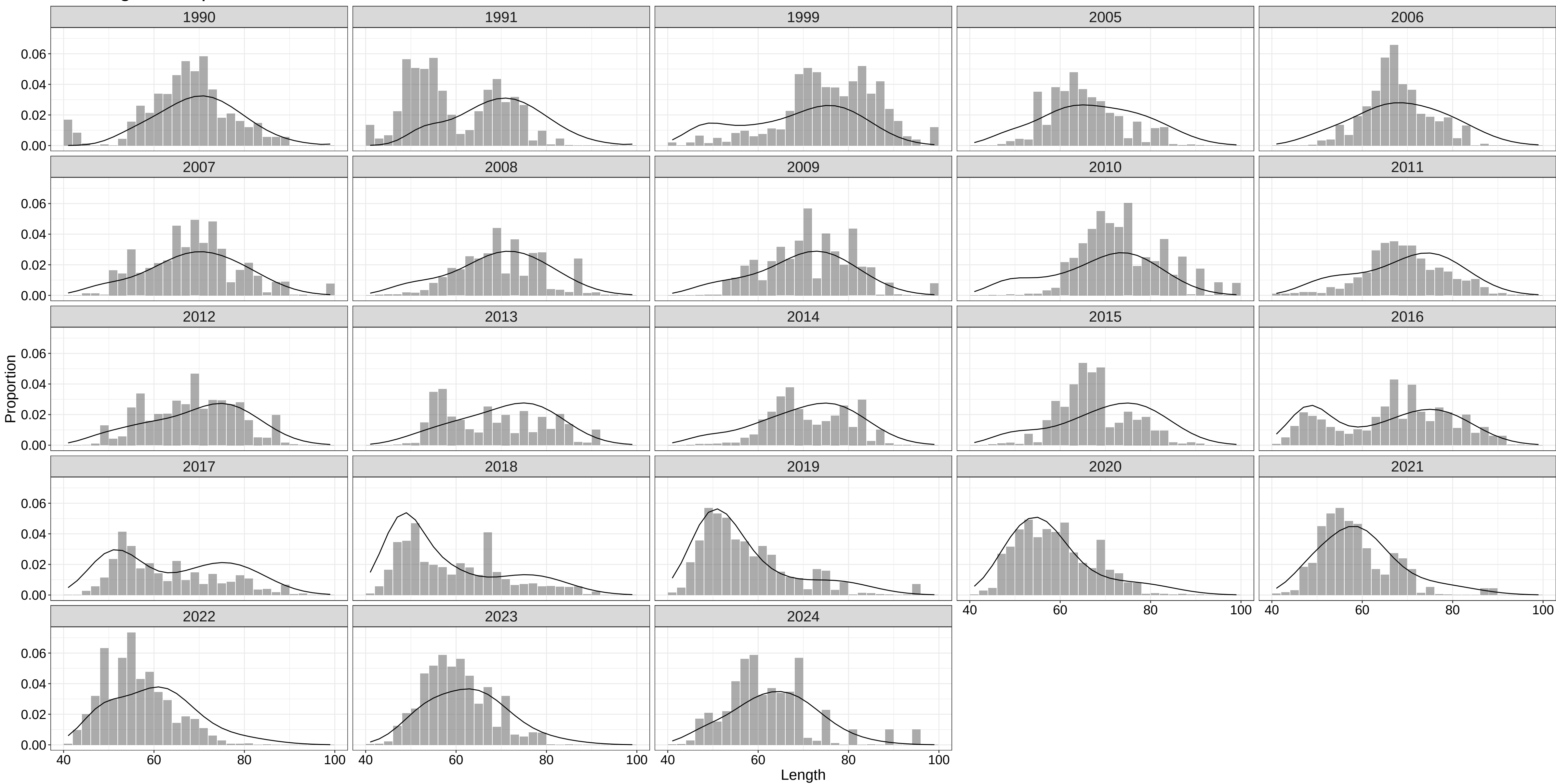
# LLF Age Comps Male



LLF Age Comps Cohort Fits Male

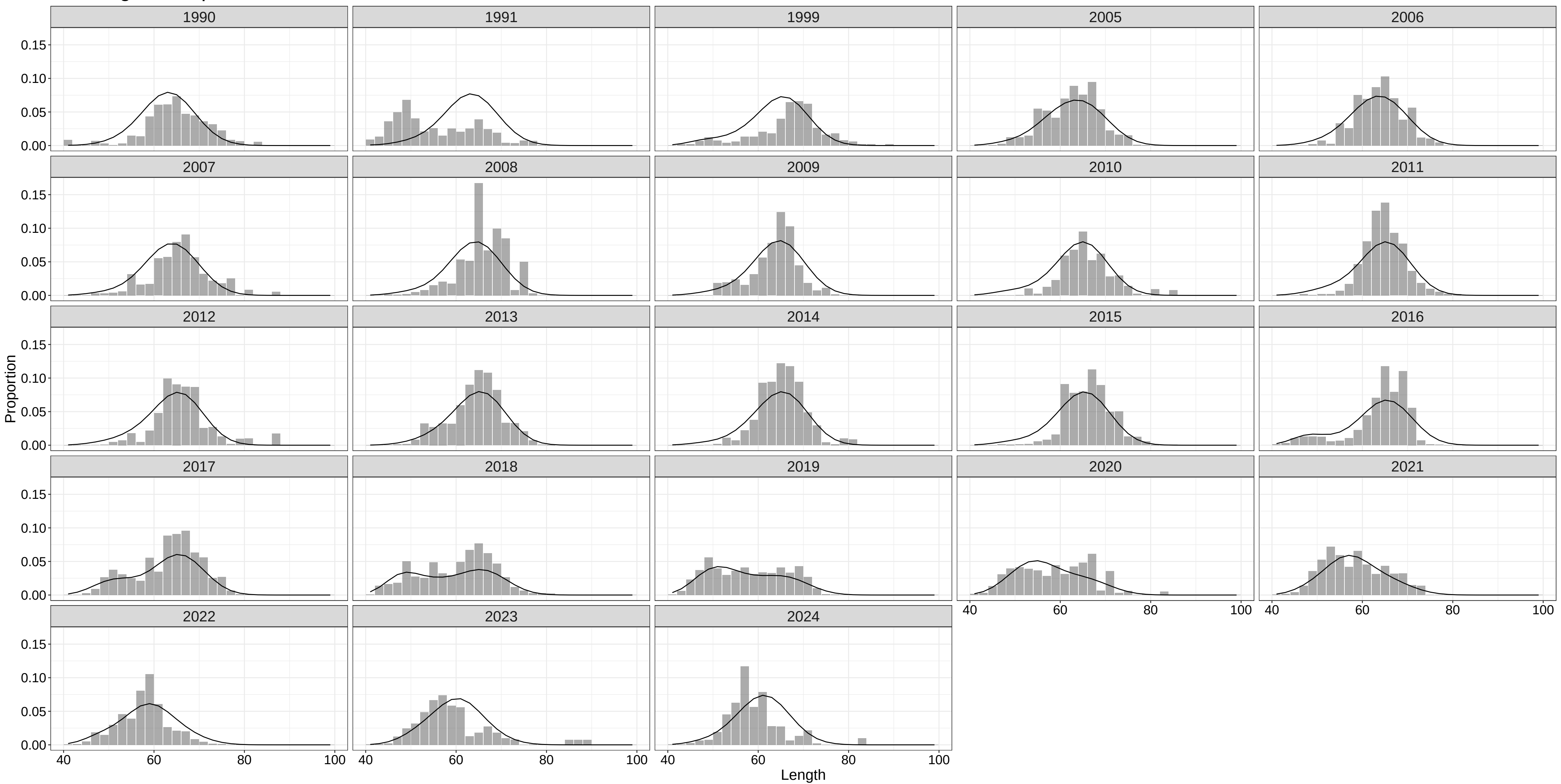


# TF Length Comps Female

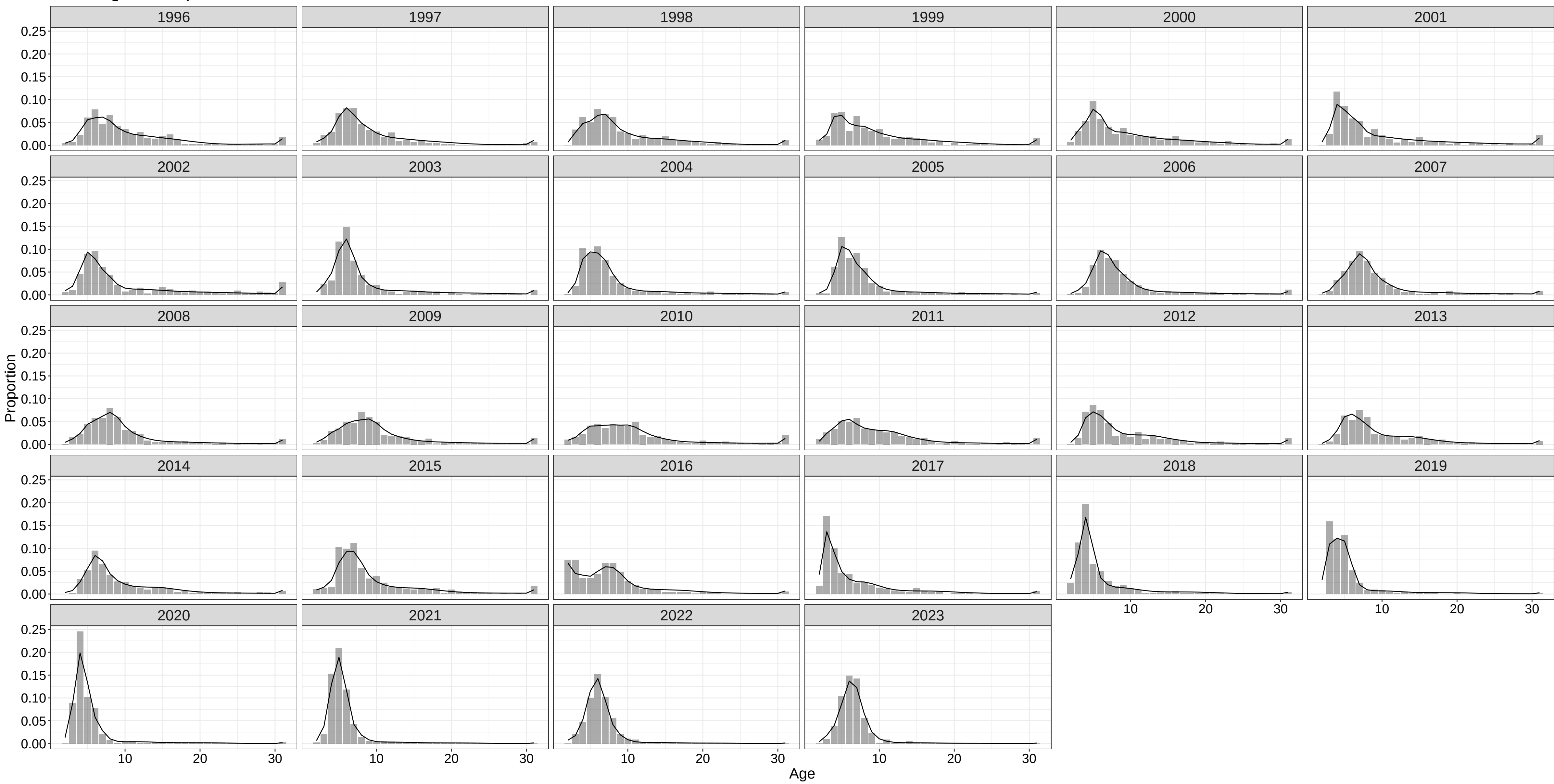




# TF Length Comps Male

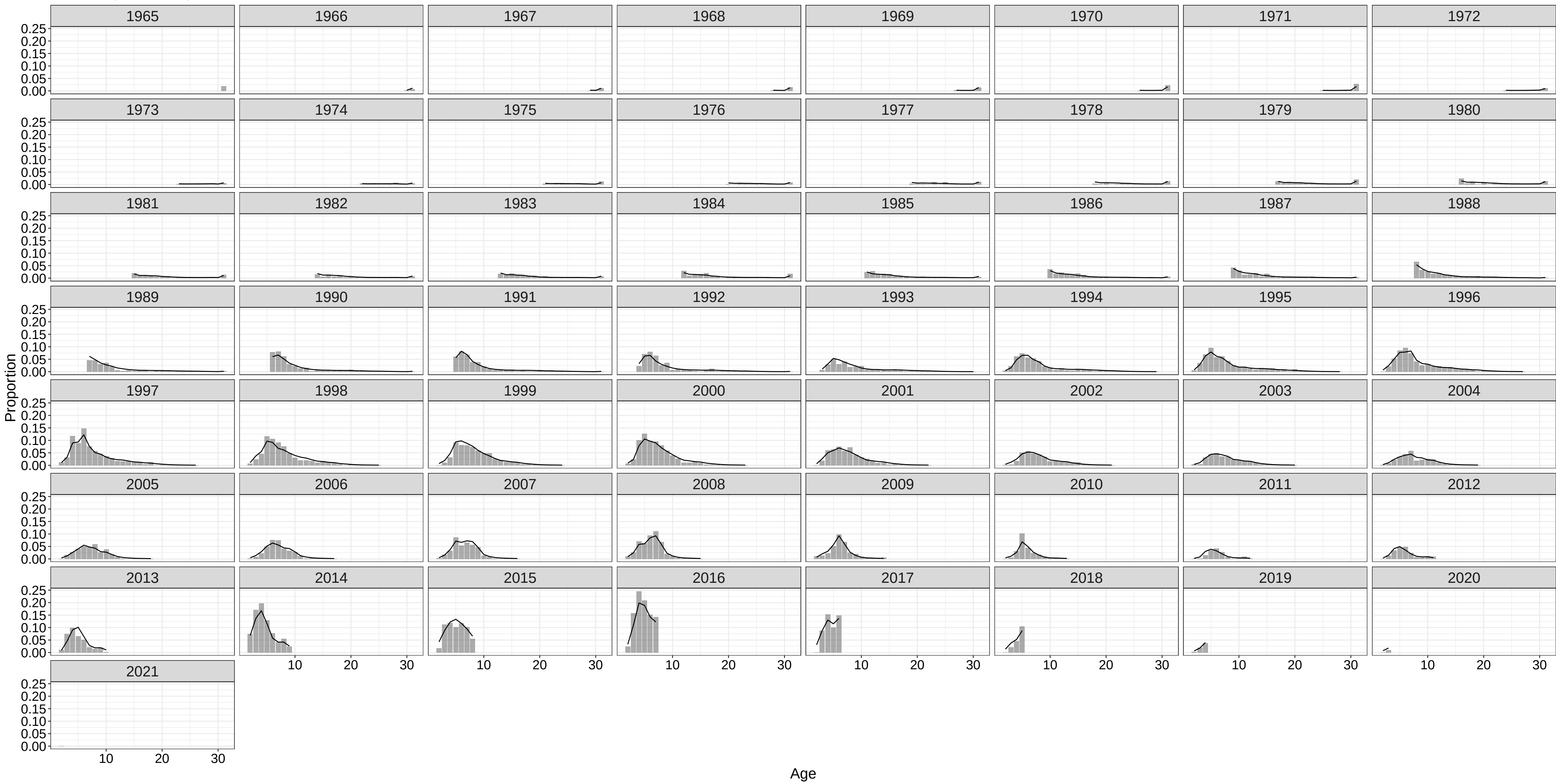


# LLS Age Comps Female

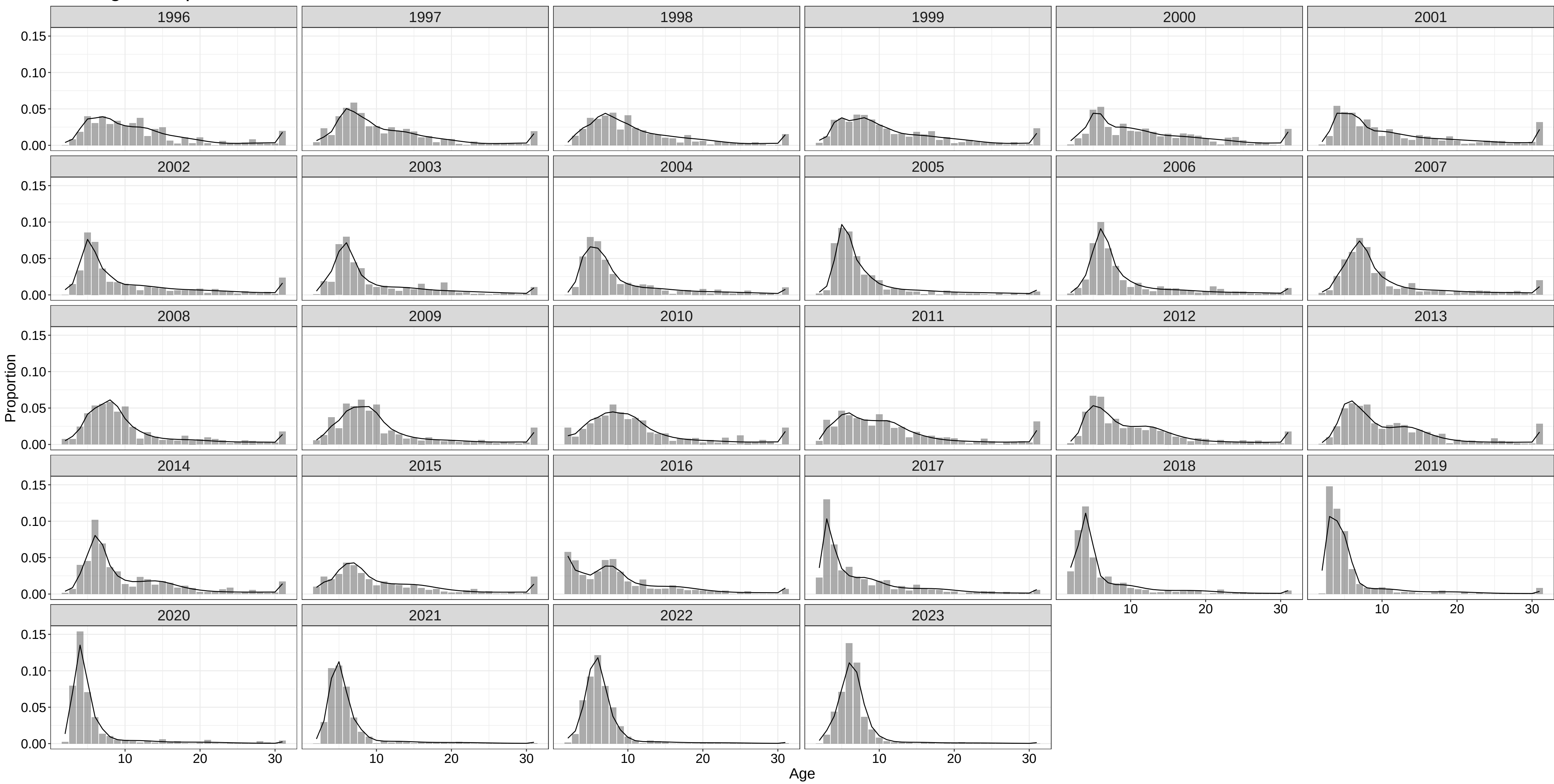




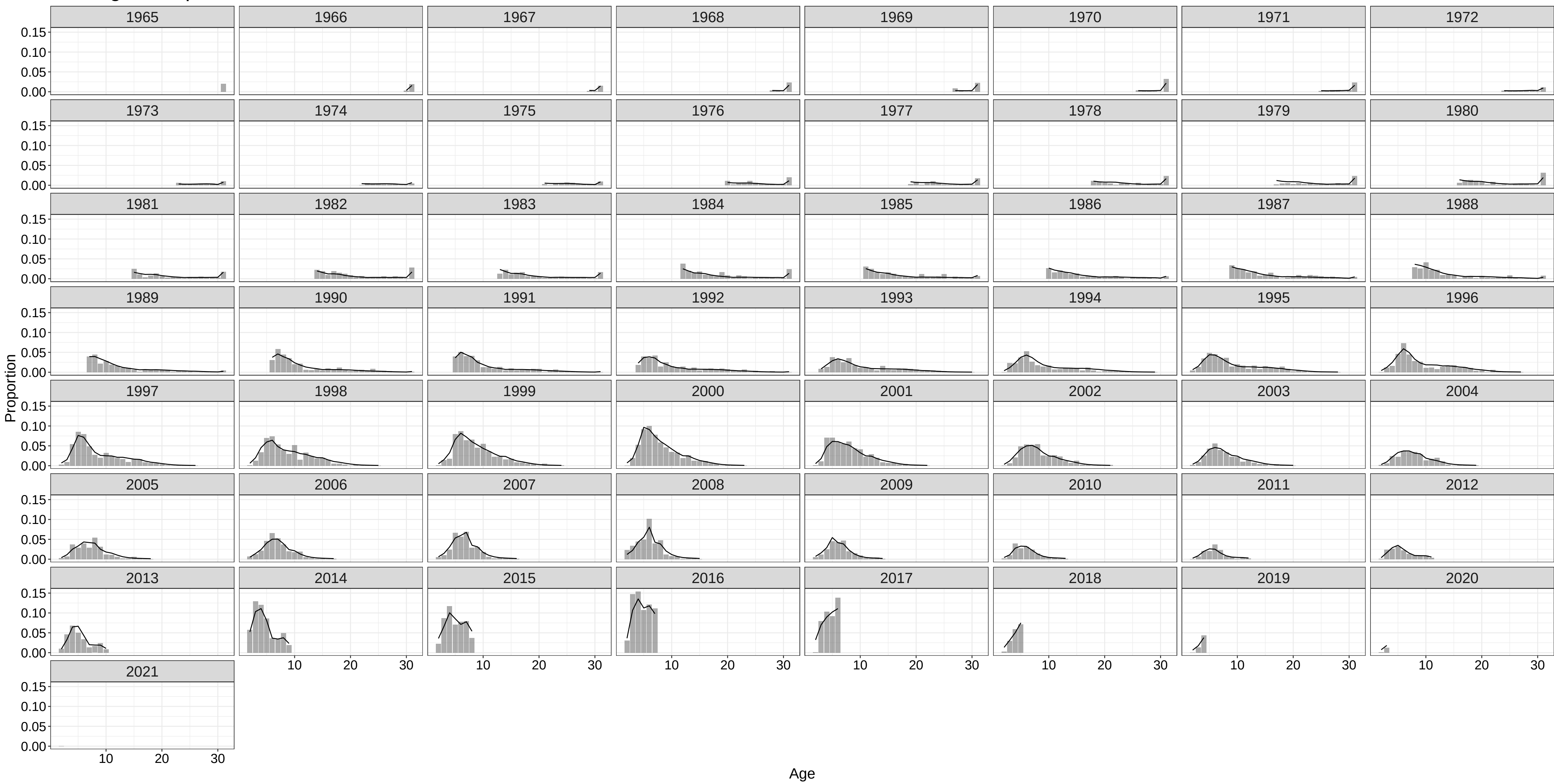
LLS Age Comps Cohort Fit Female



# LLS Age Comps Male

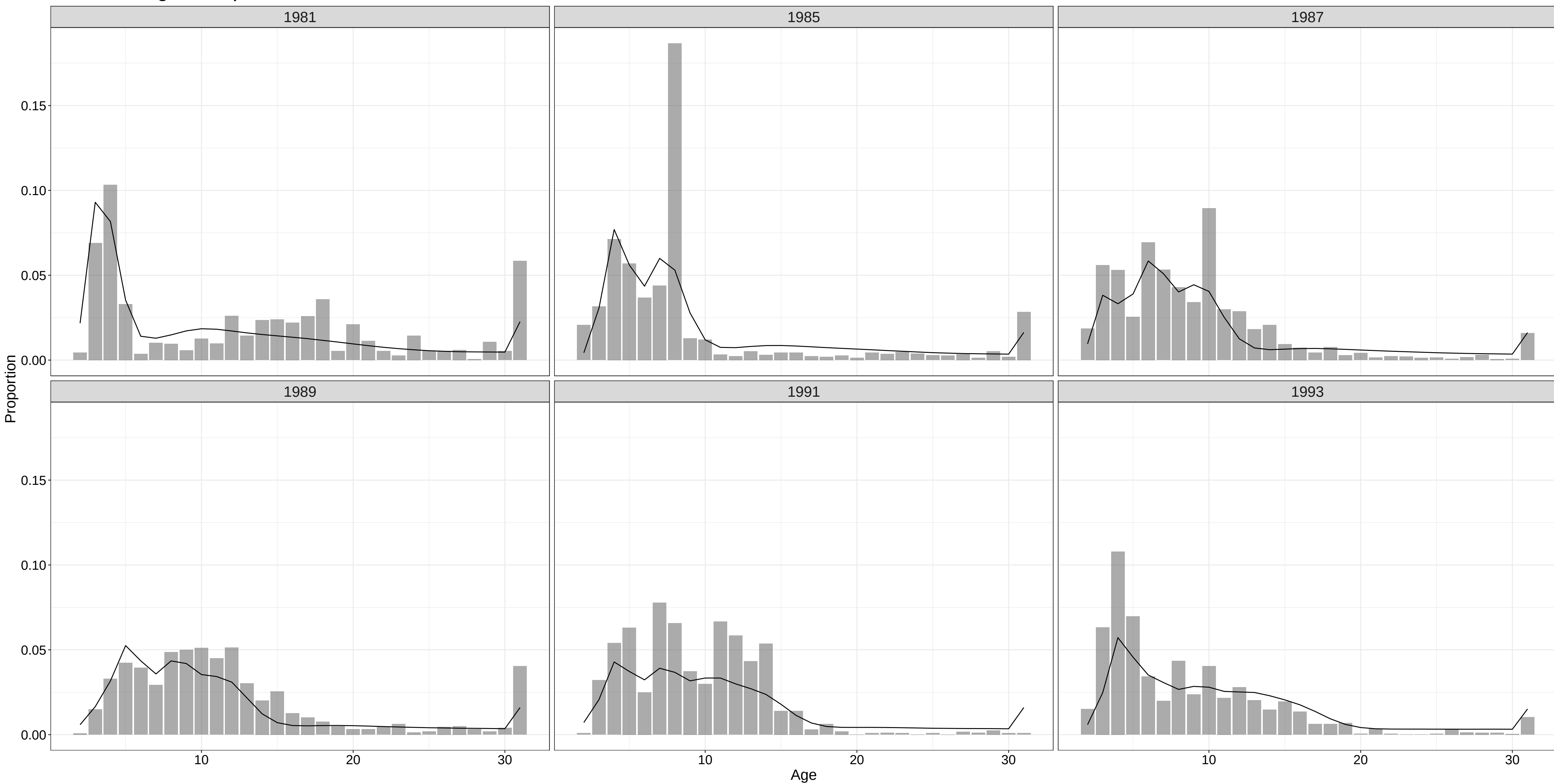


# LLS Age Comps Cohort Fit Male

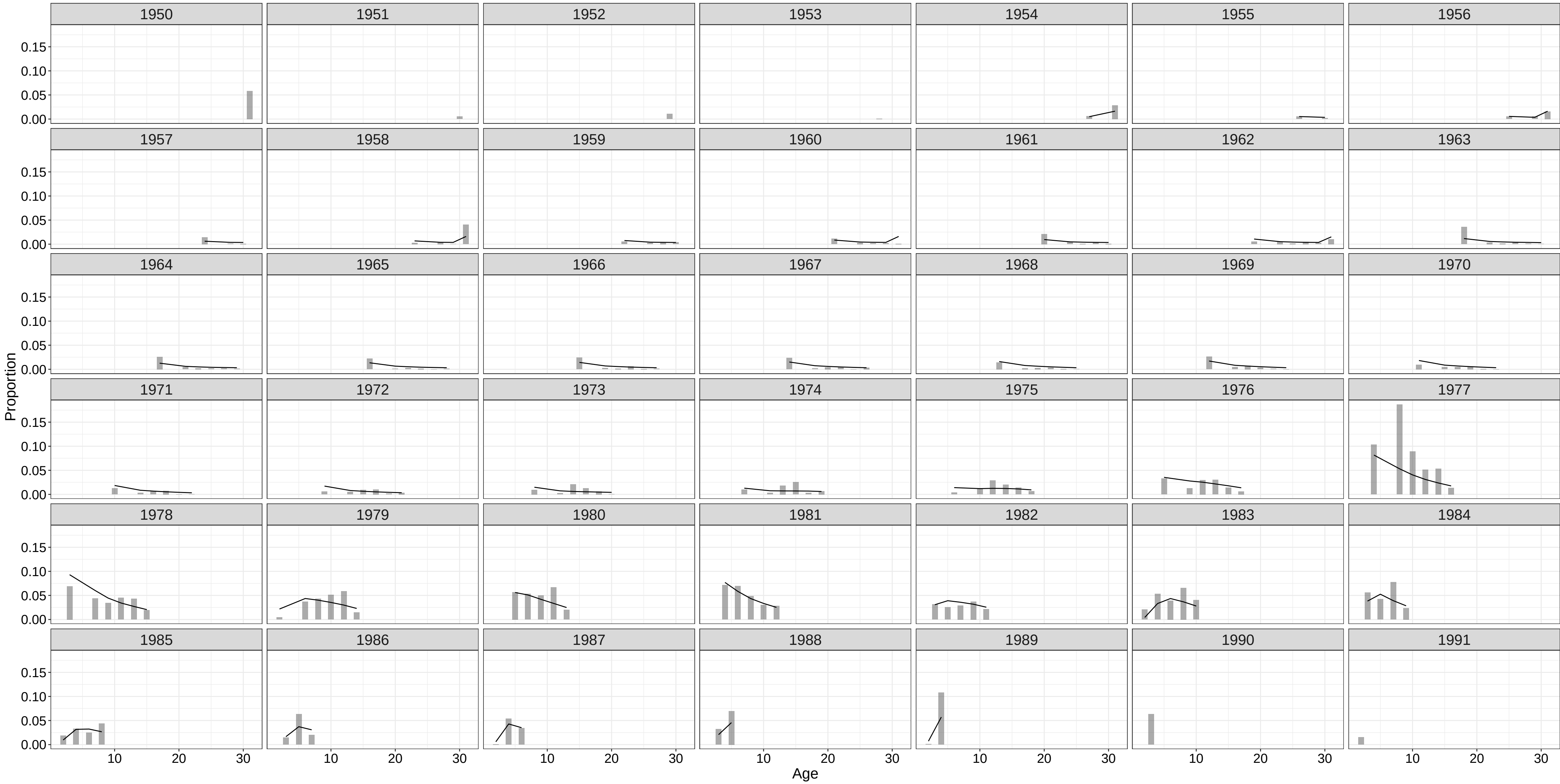




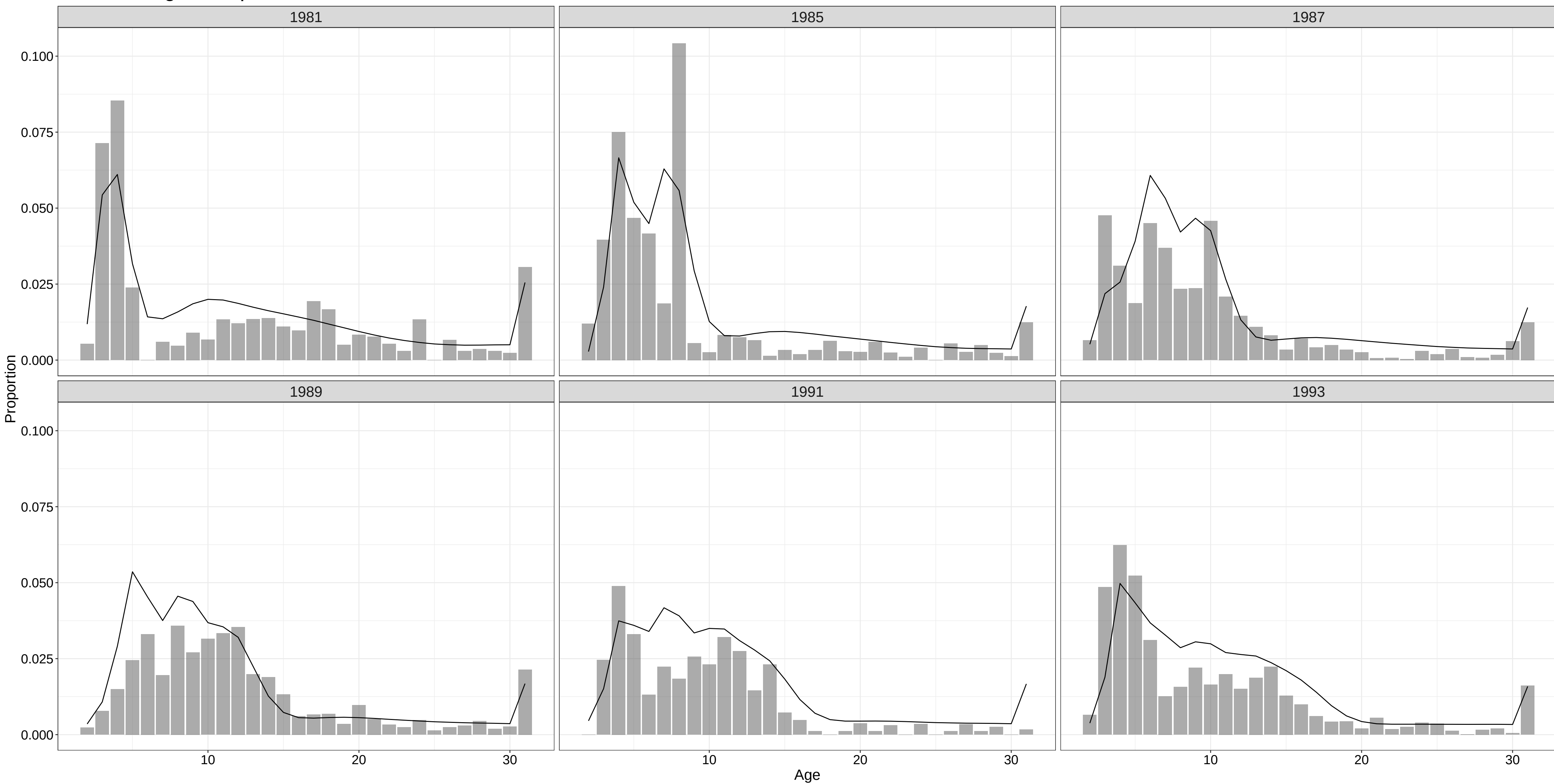
# JPN LLS Age Comps Female



JPN LLS Age Comps Cohort Fit Female

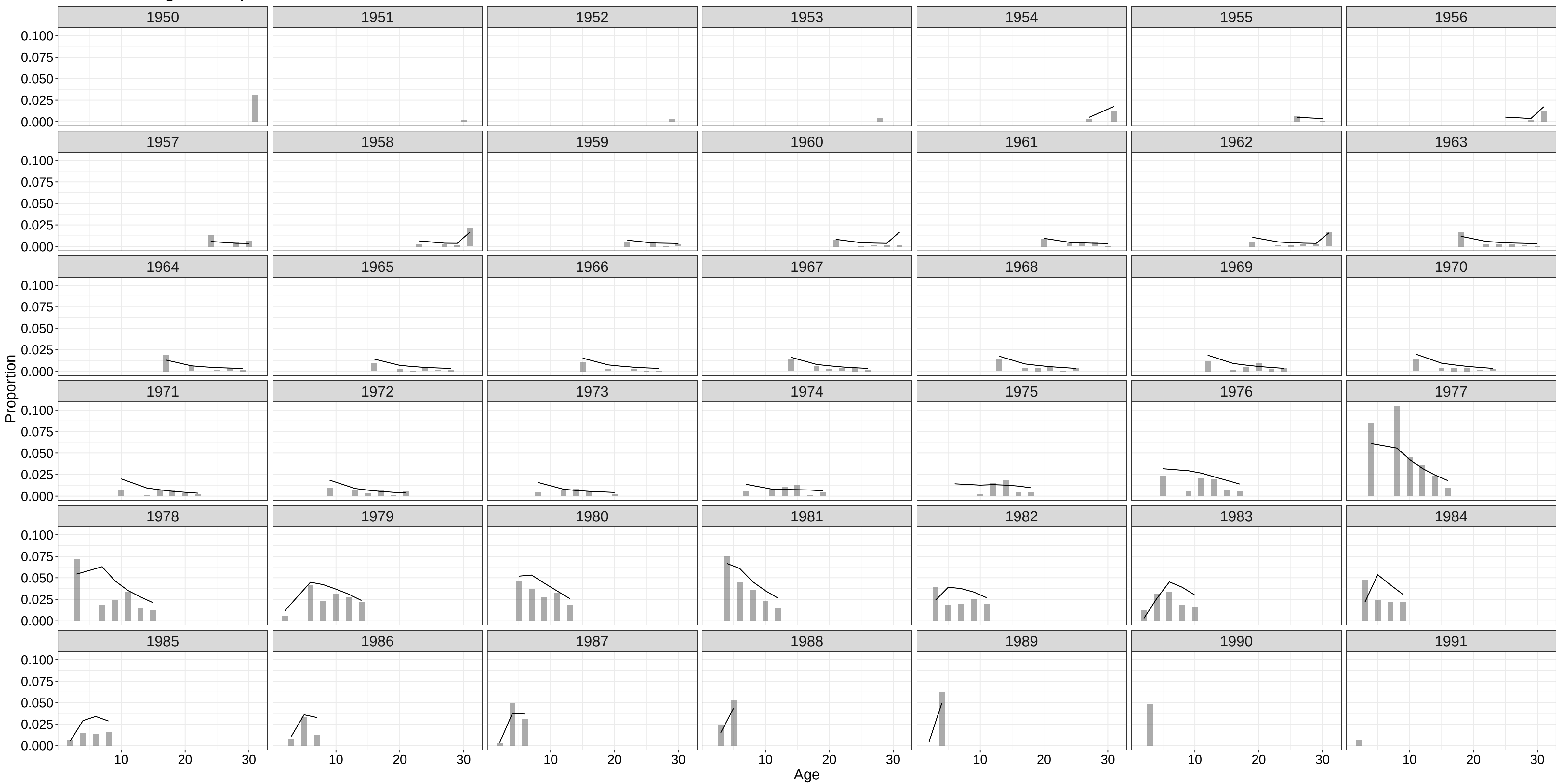


# JPN LLS Age Comps Male

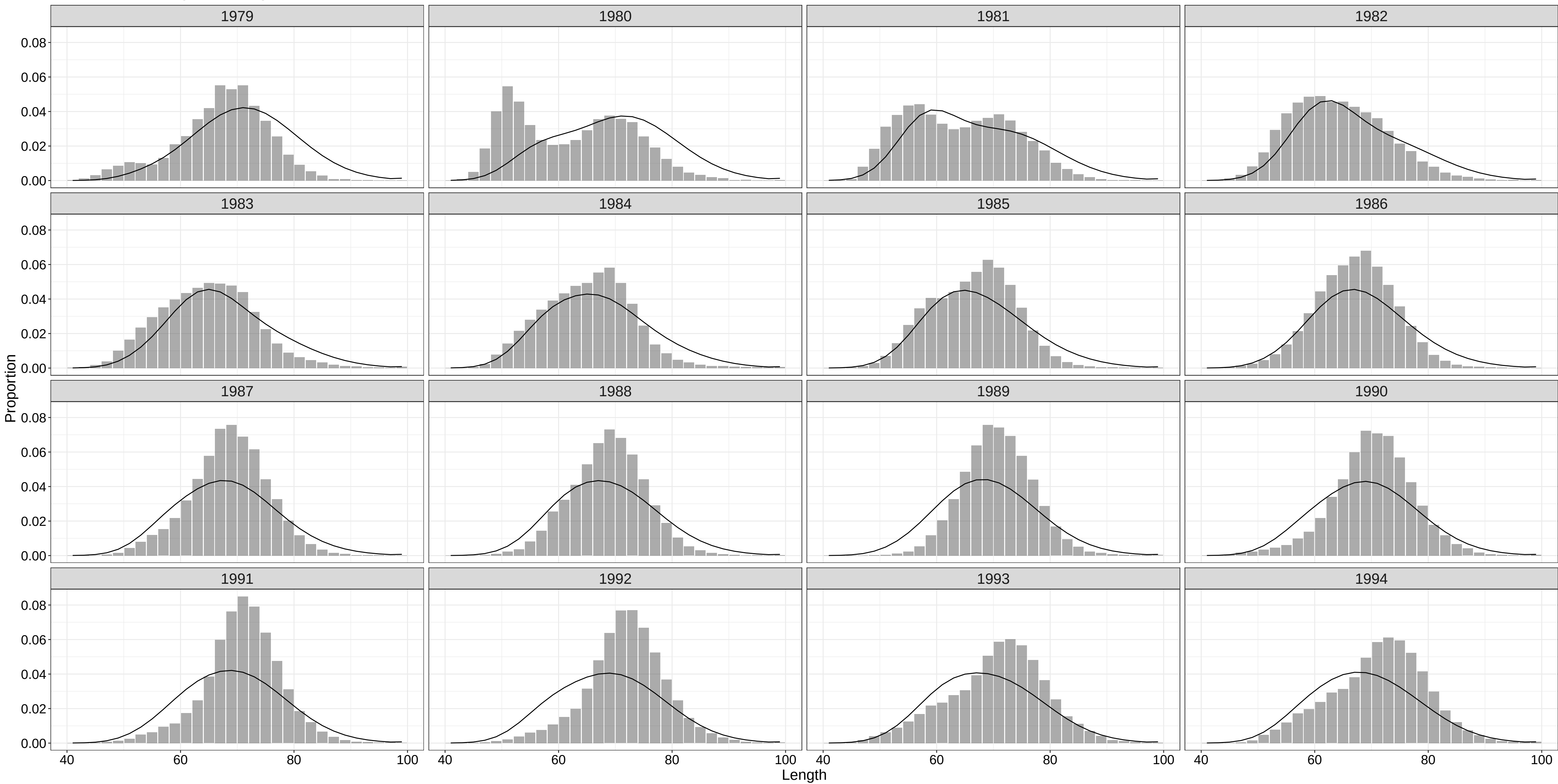




JPN LLS Age Comps Cohort Fit Male

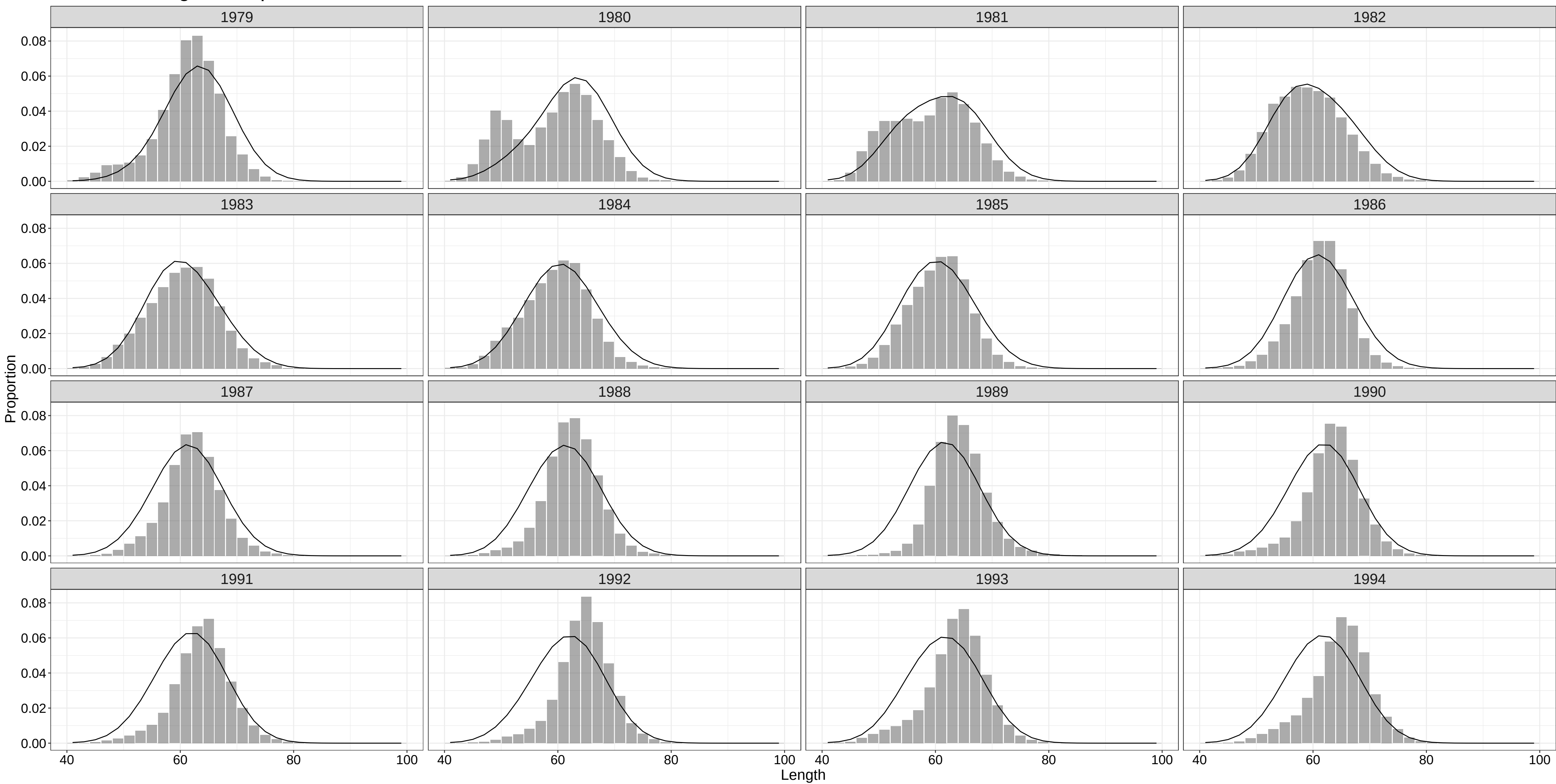


# JPN LLS Length Comps Female

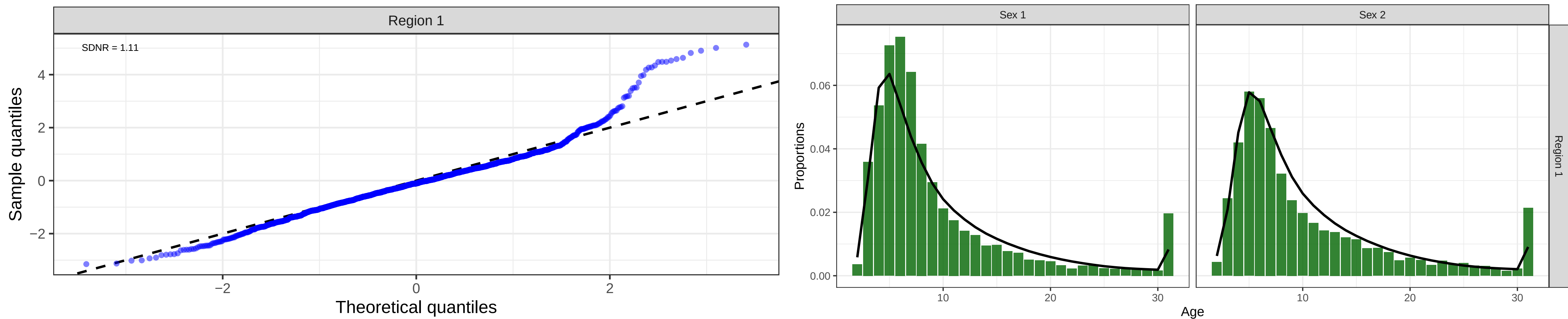
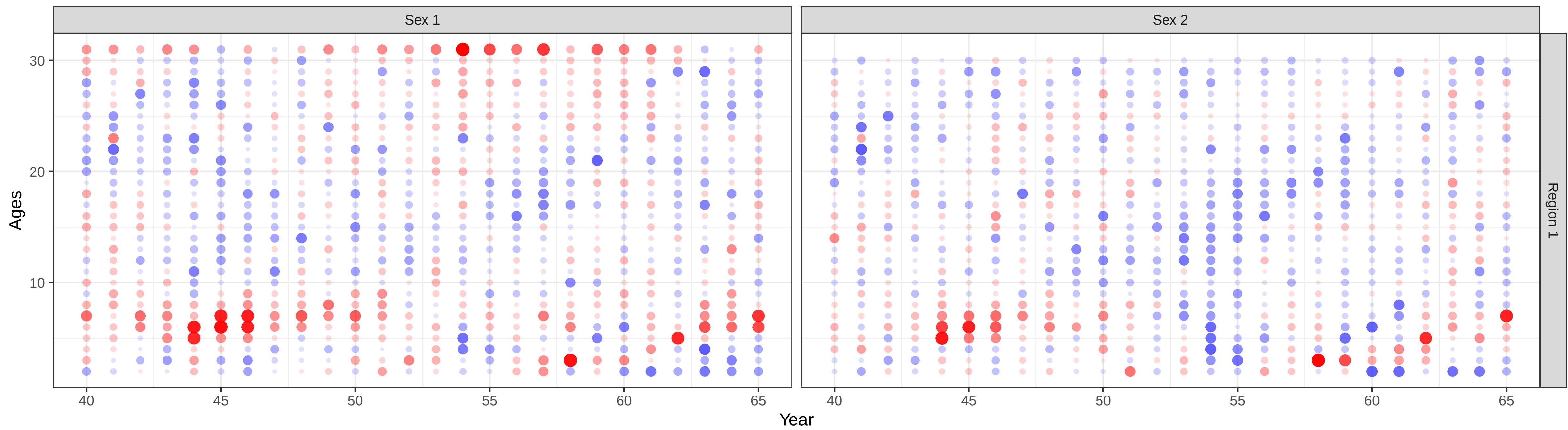


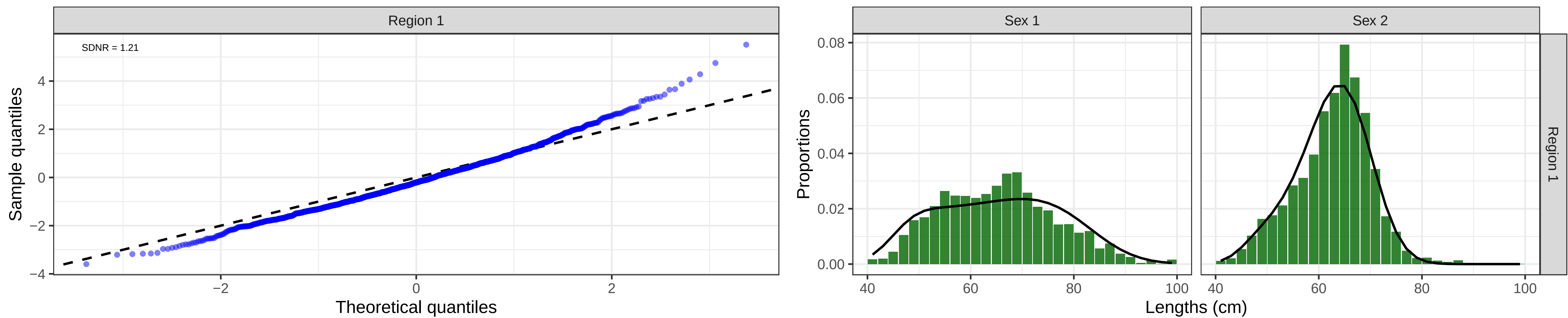
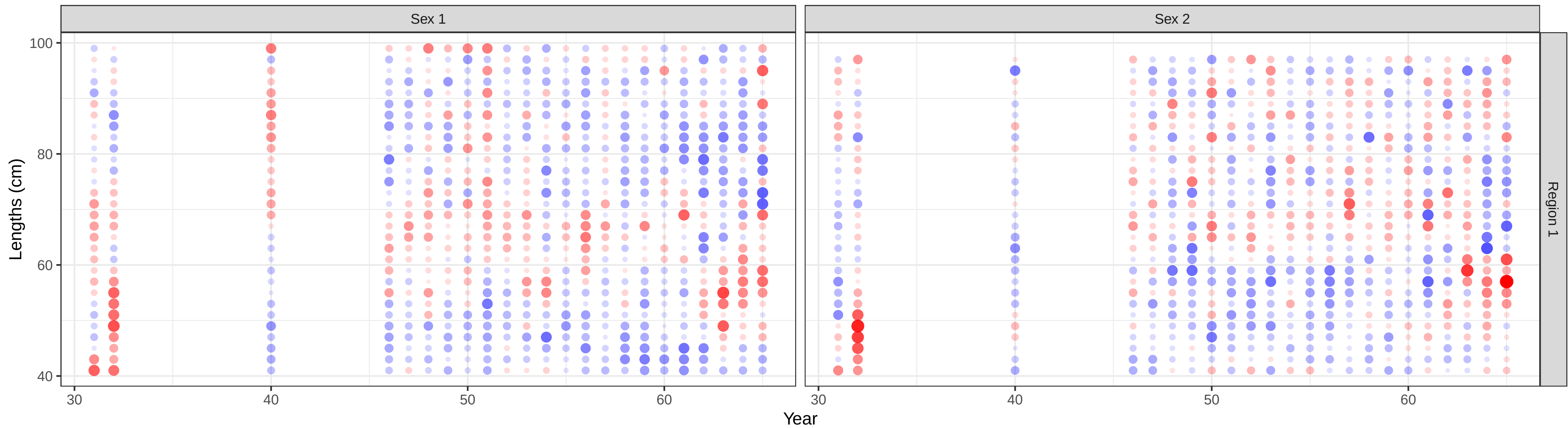


JPN LLS Length Comps Male

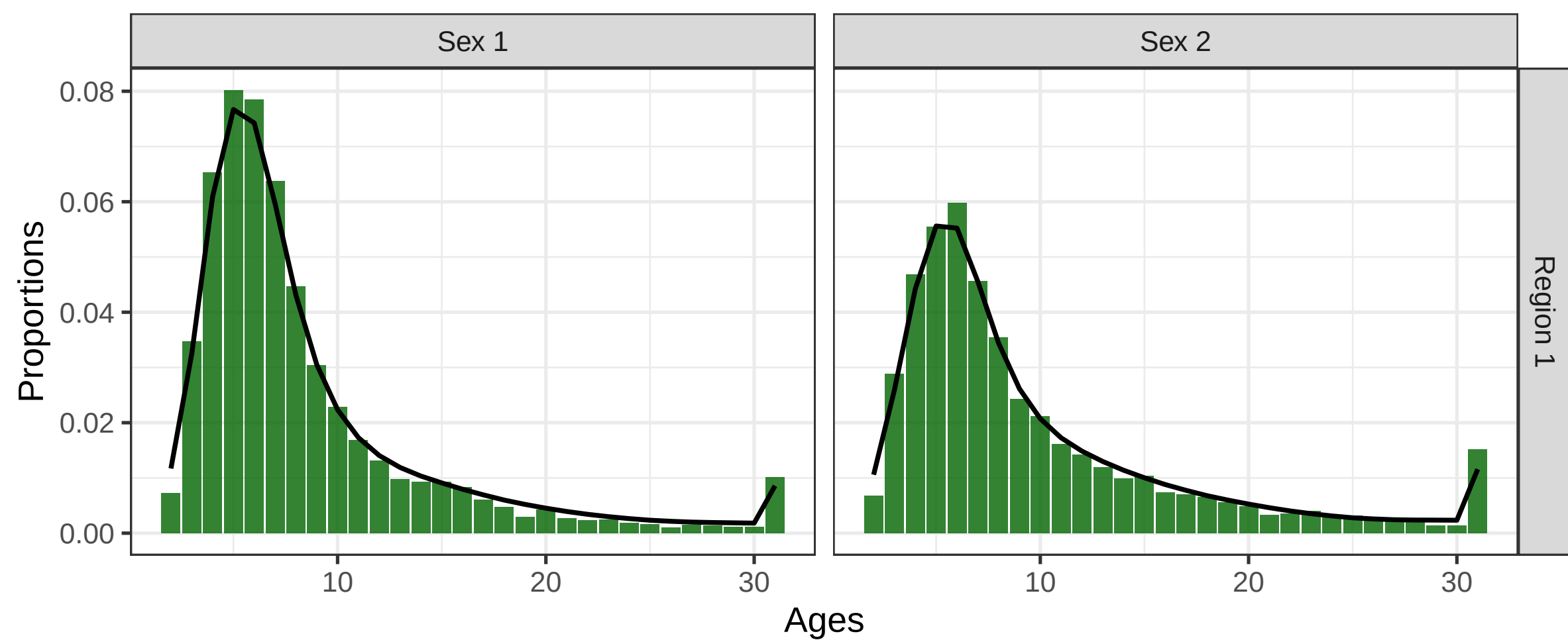
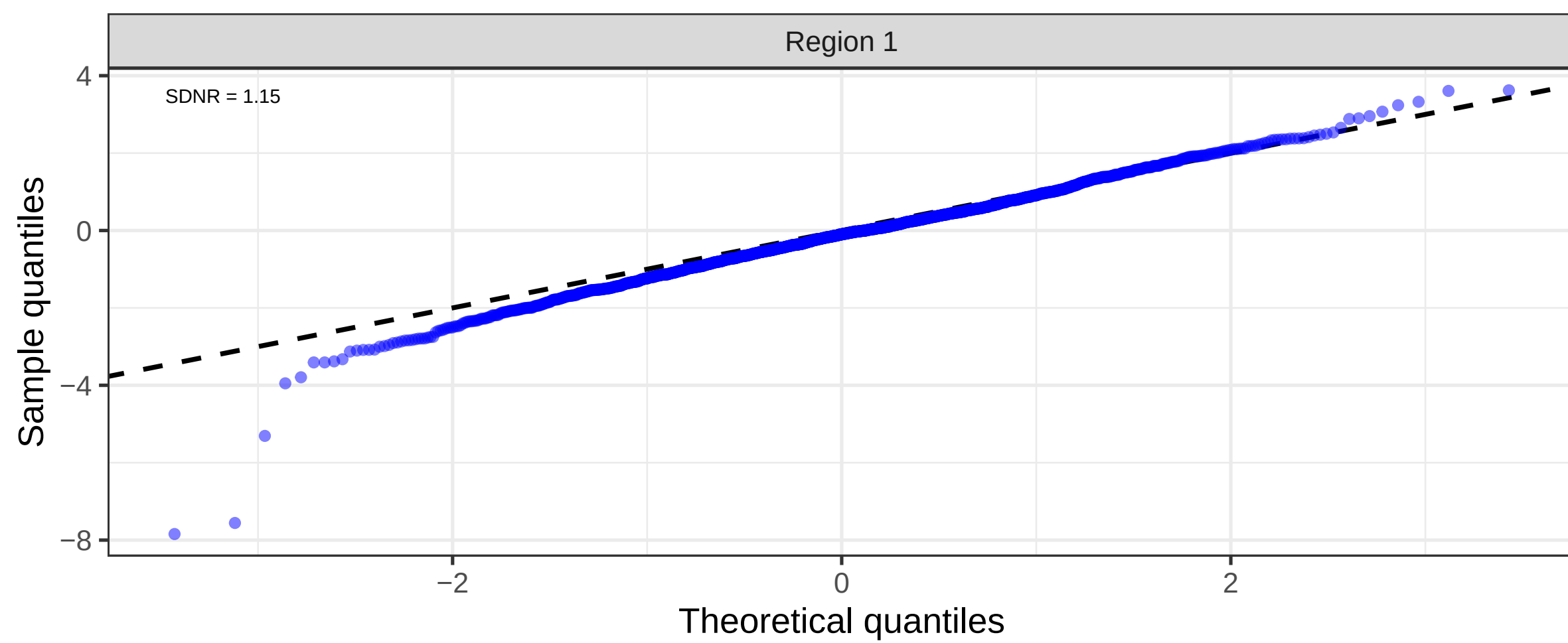
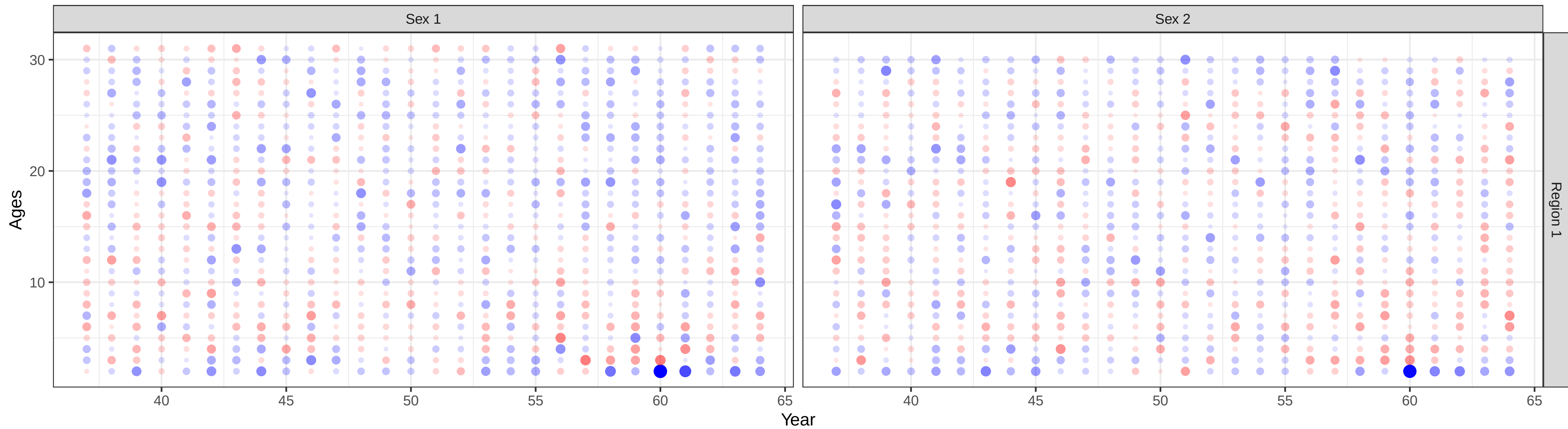


Sign   ● Neg   ● Pos   abs(Resid)   ● 1   ● 2   ● 3   ● 4   ● 5

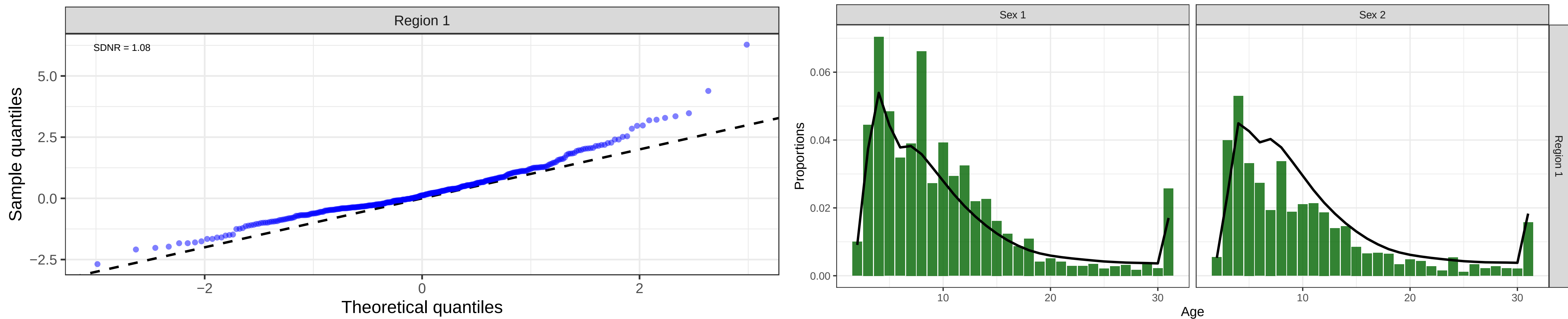
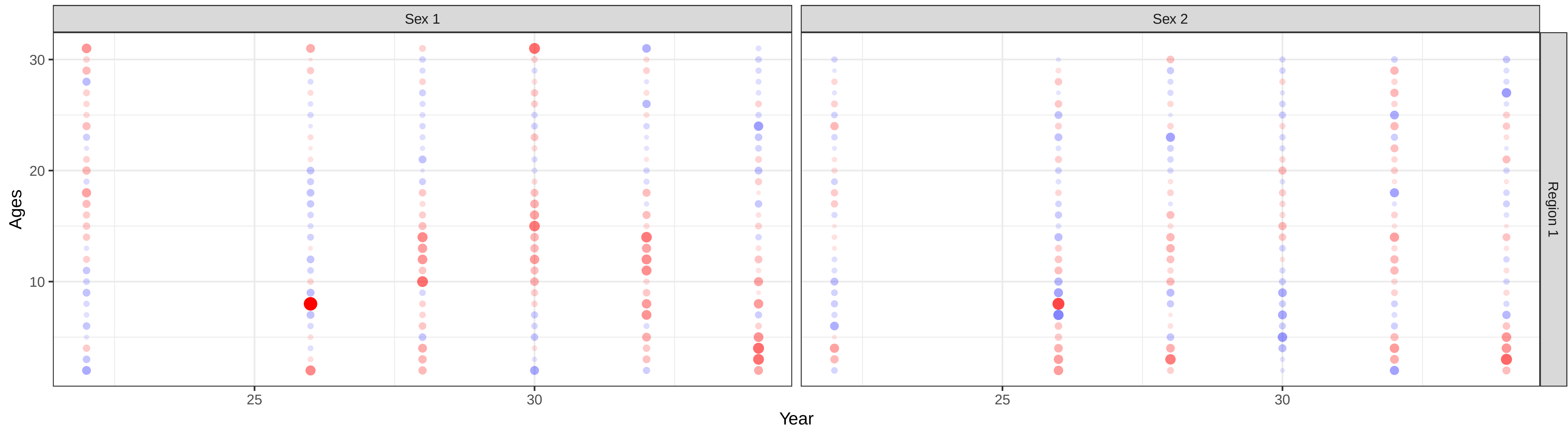




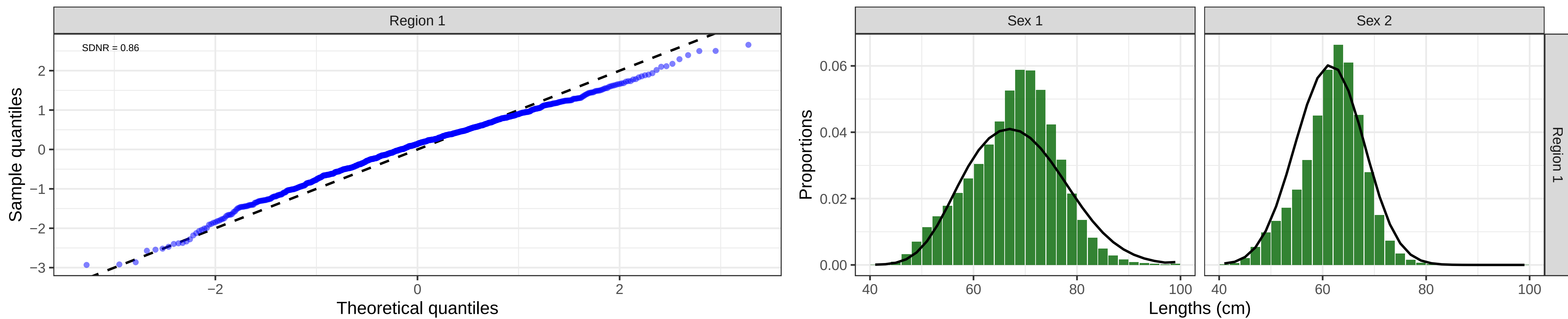
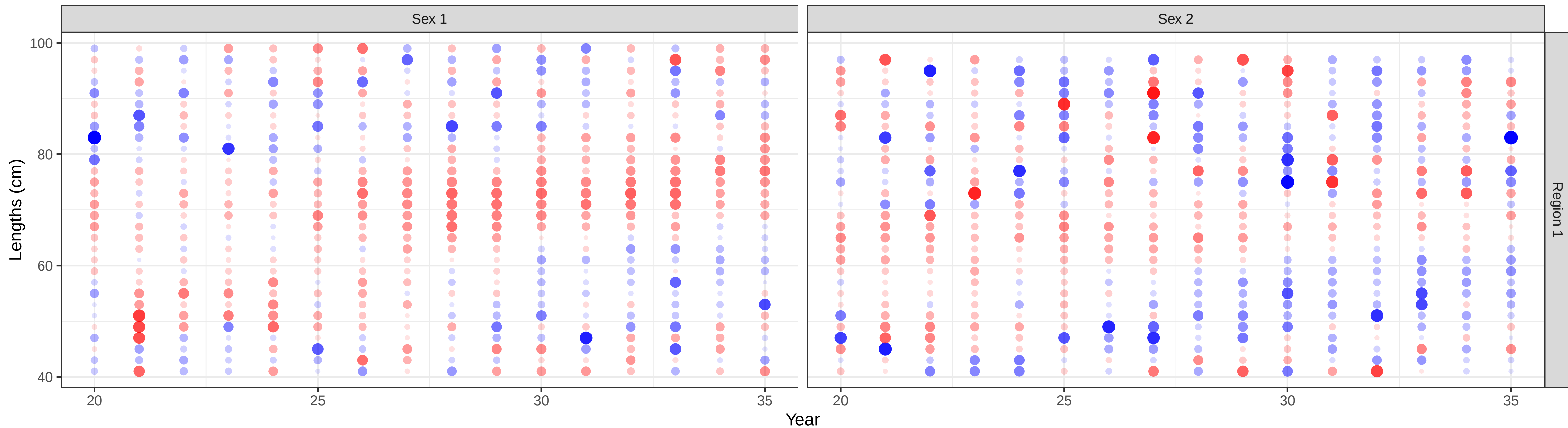




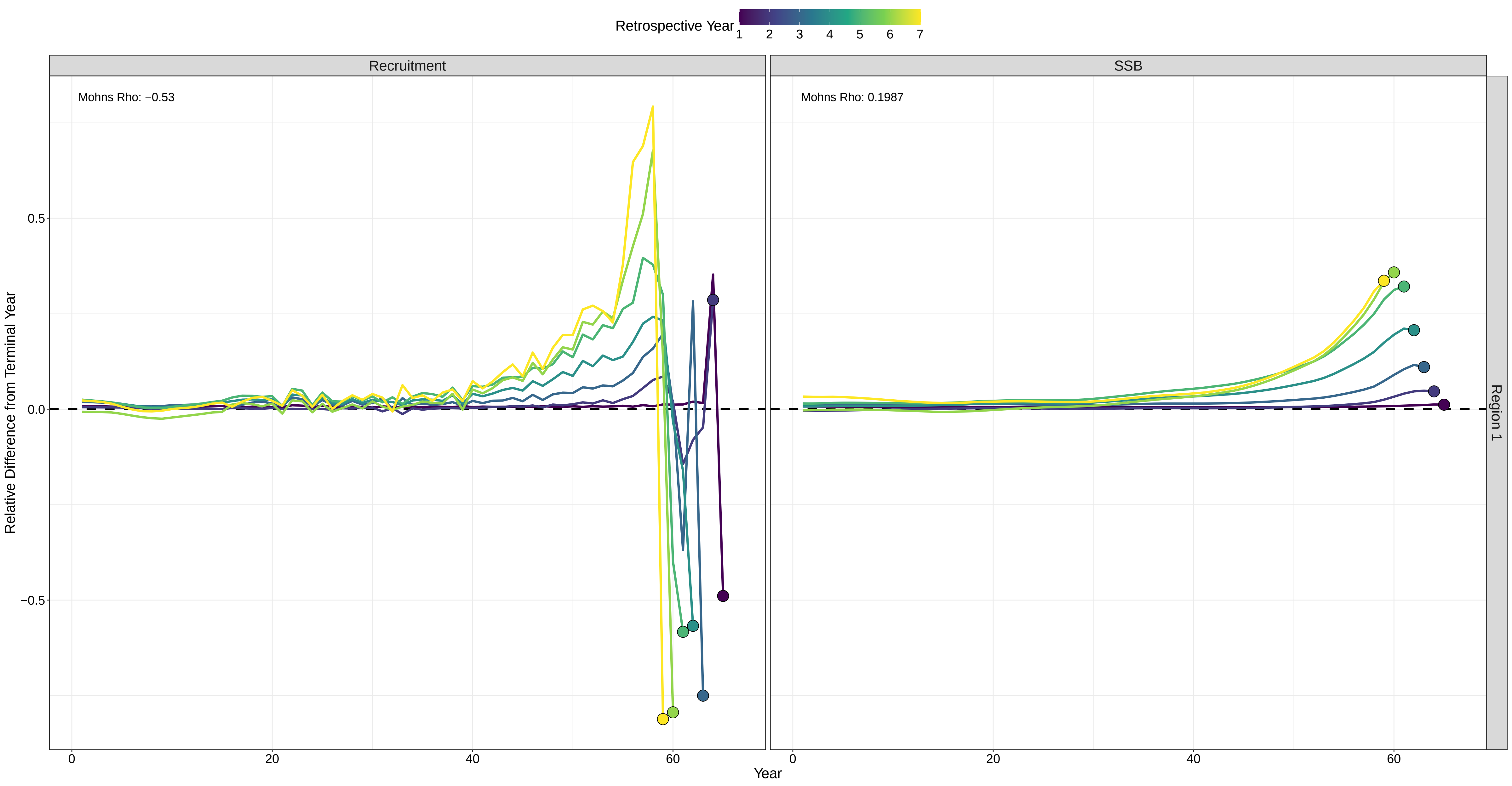
Sign   ● Neg   ● Pos   abs(Resid)   ● 2   ● 4   ● 6



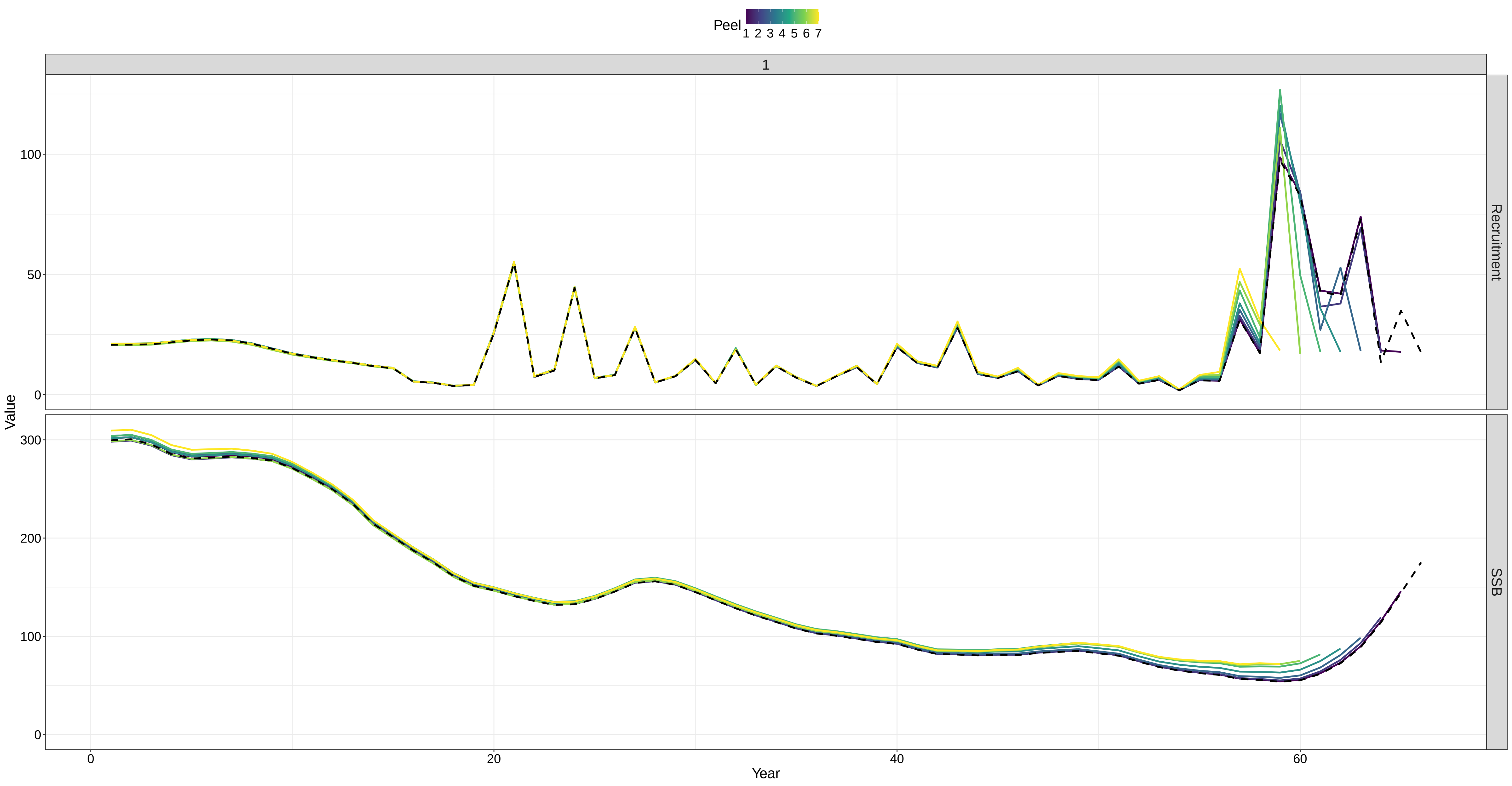
Sign   ● Neg   ● Pos   abs(Resid)   ● 1   ● 2



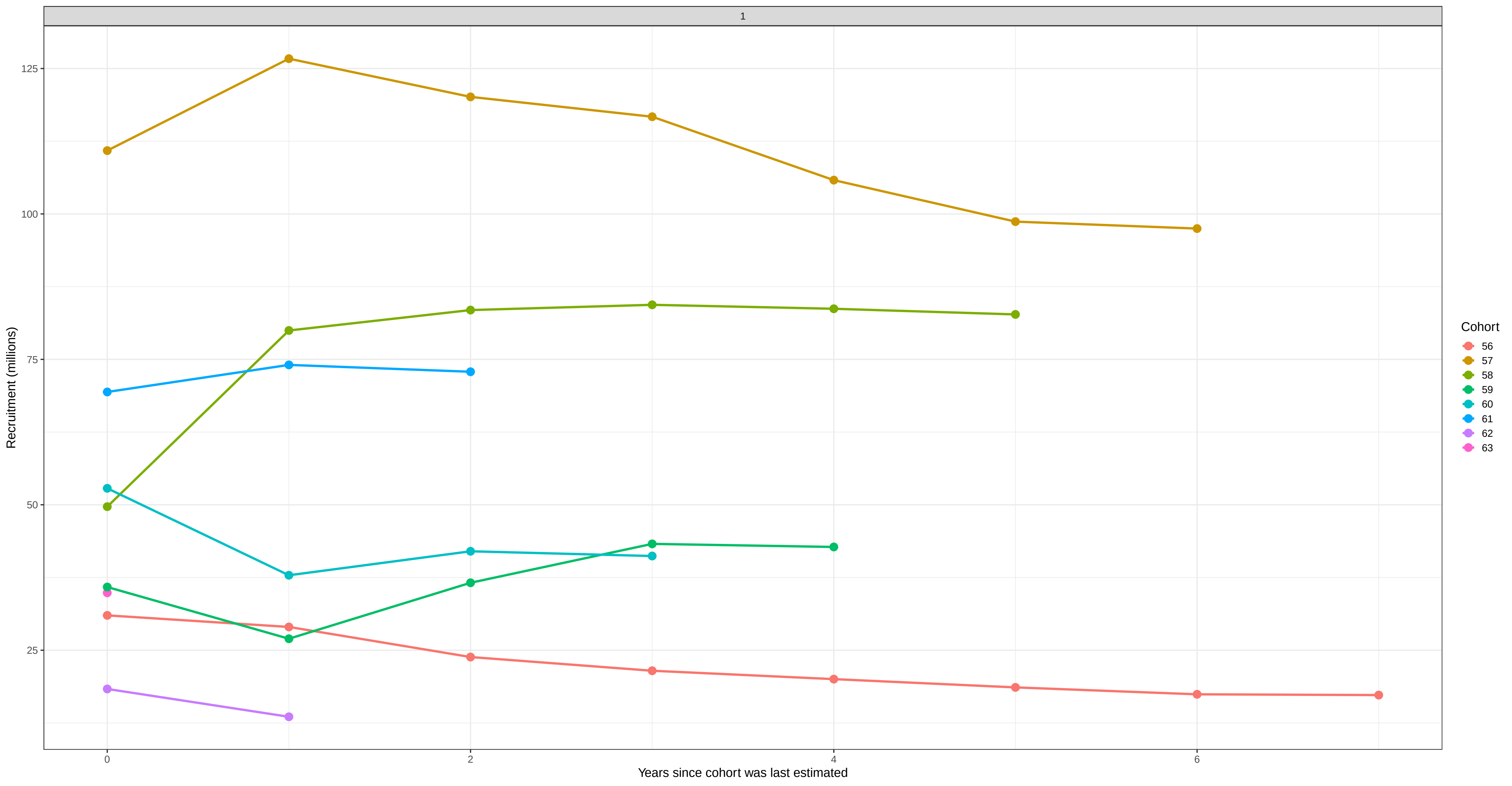




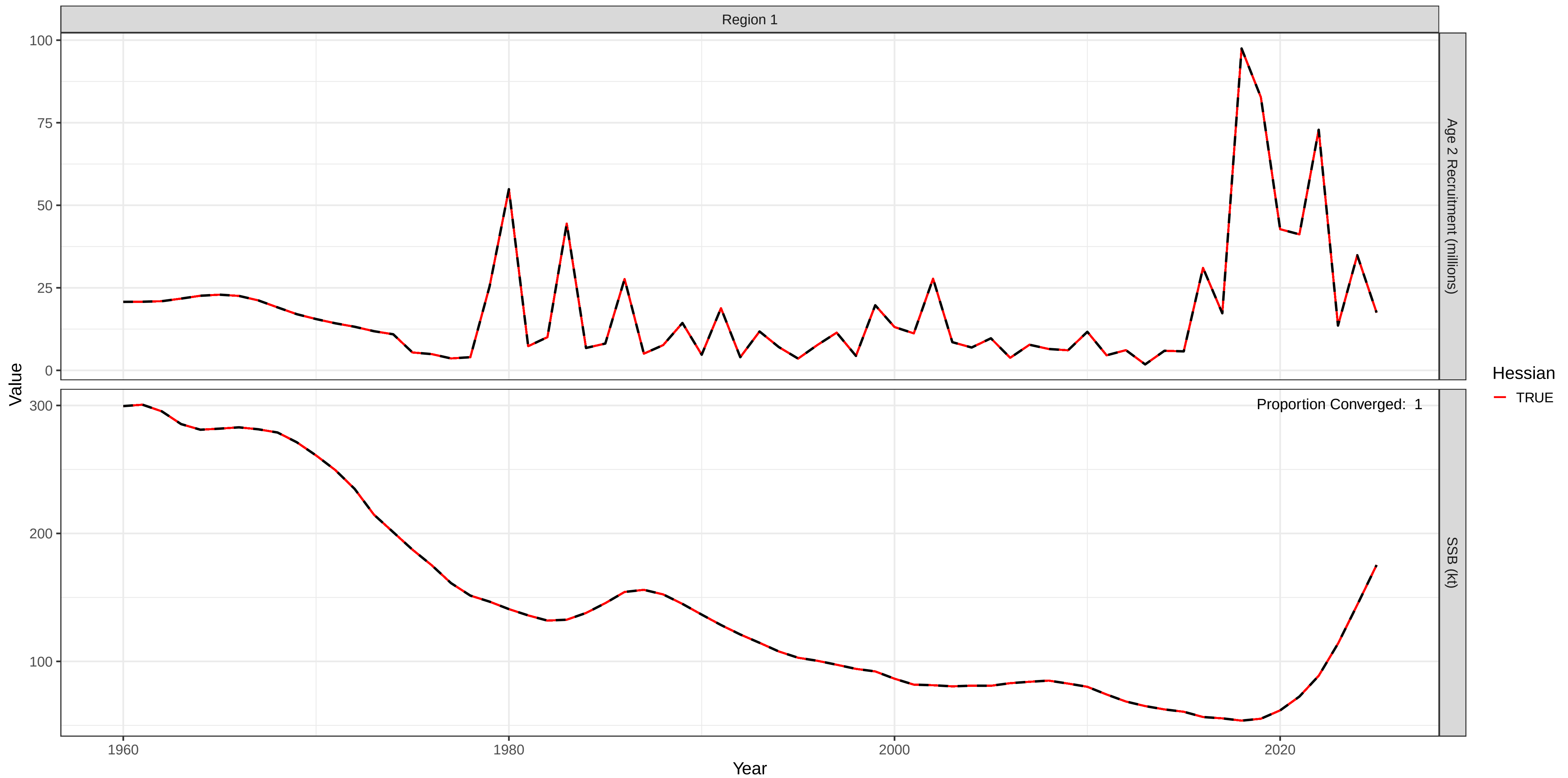




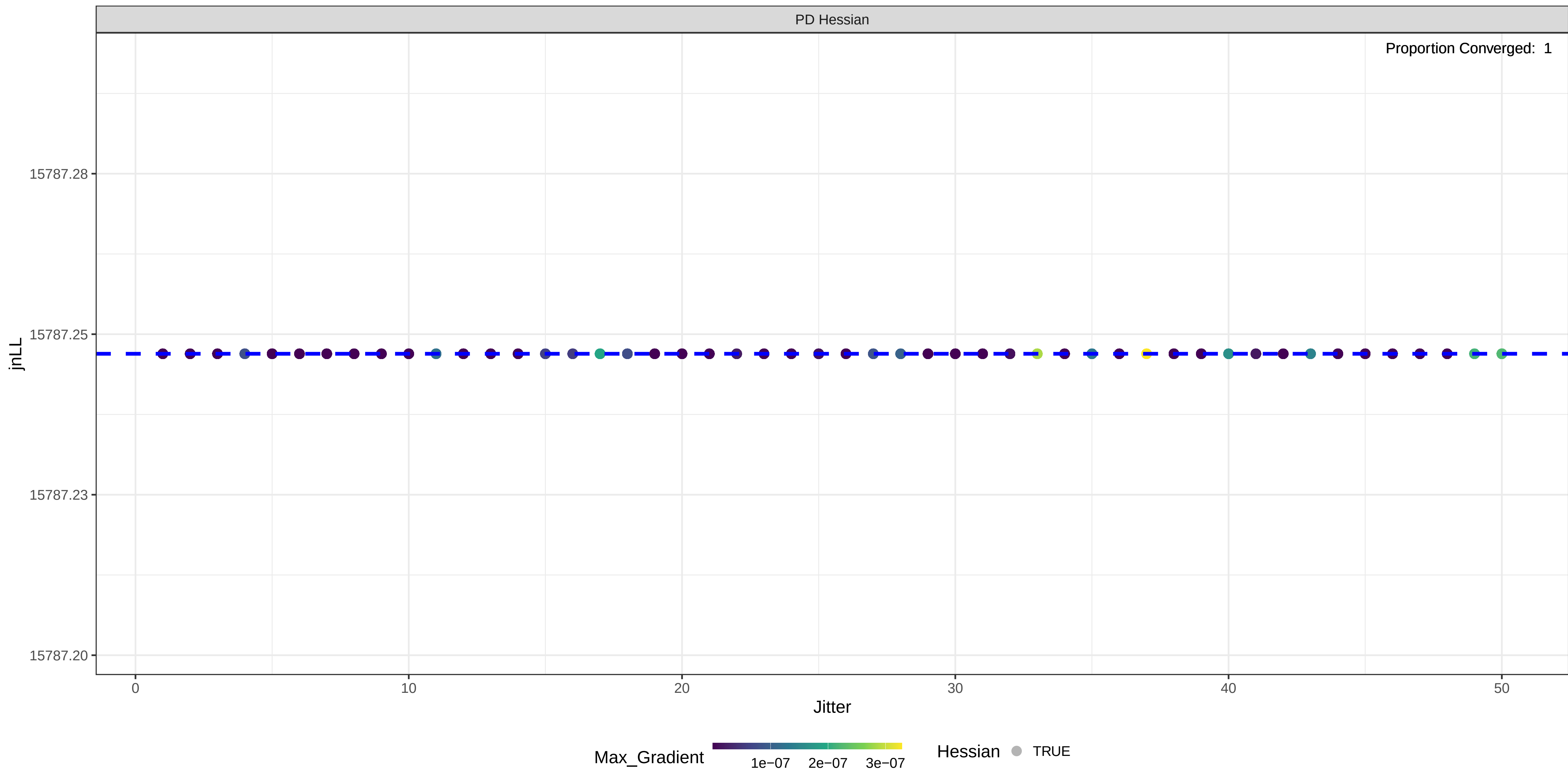




# Jitter SSB and Recruitment

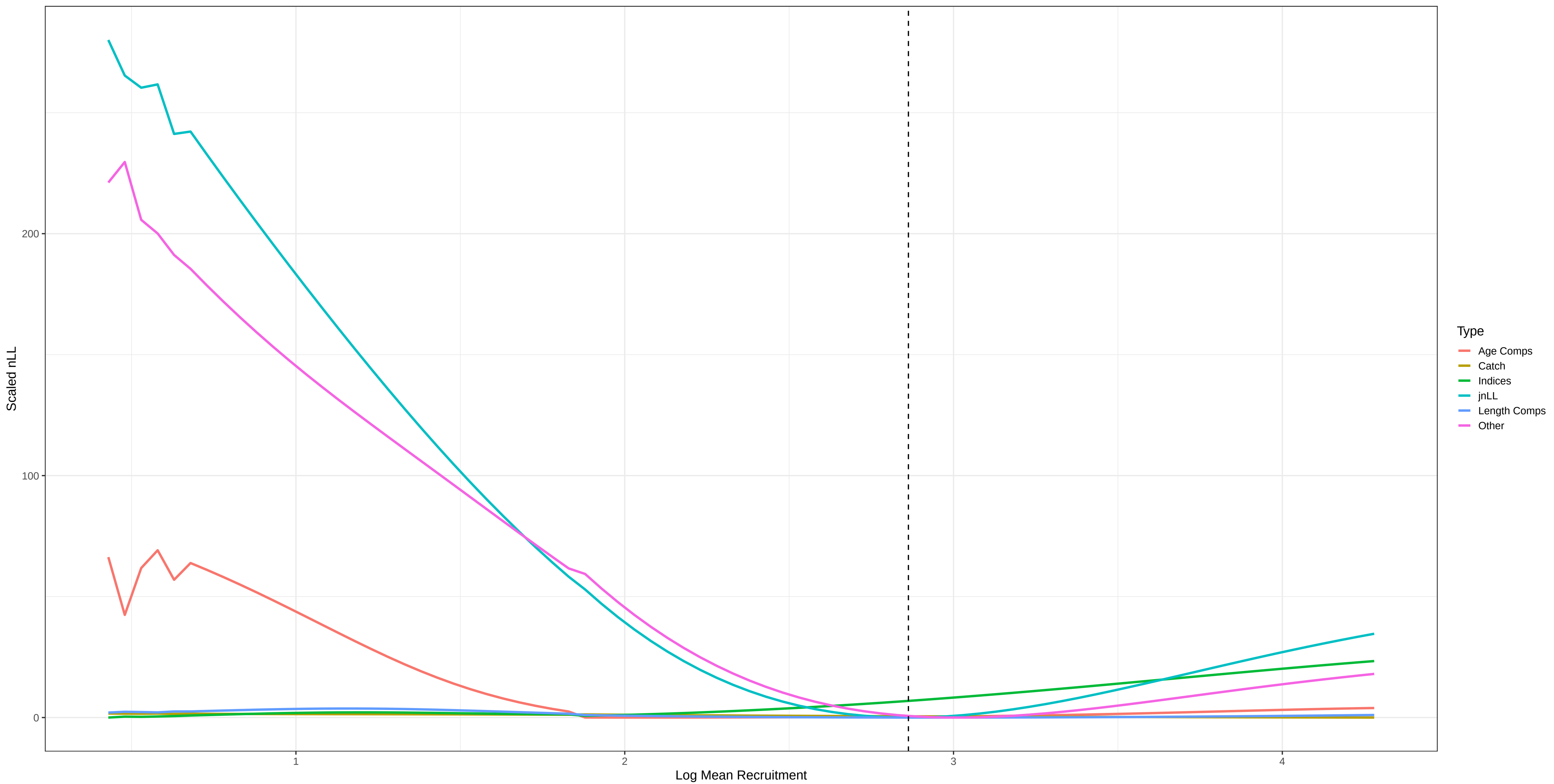


Jitter nLL and Convergence

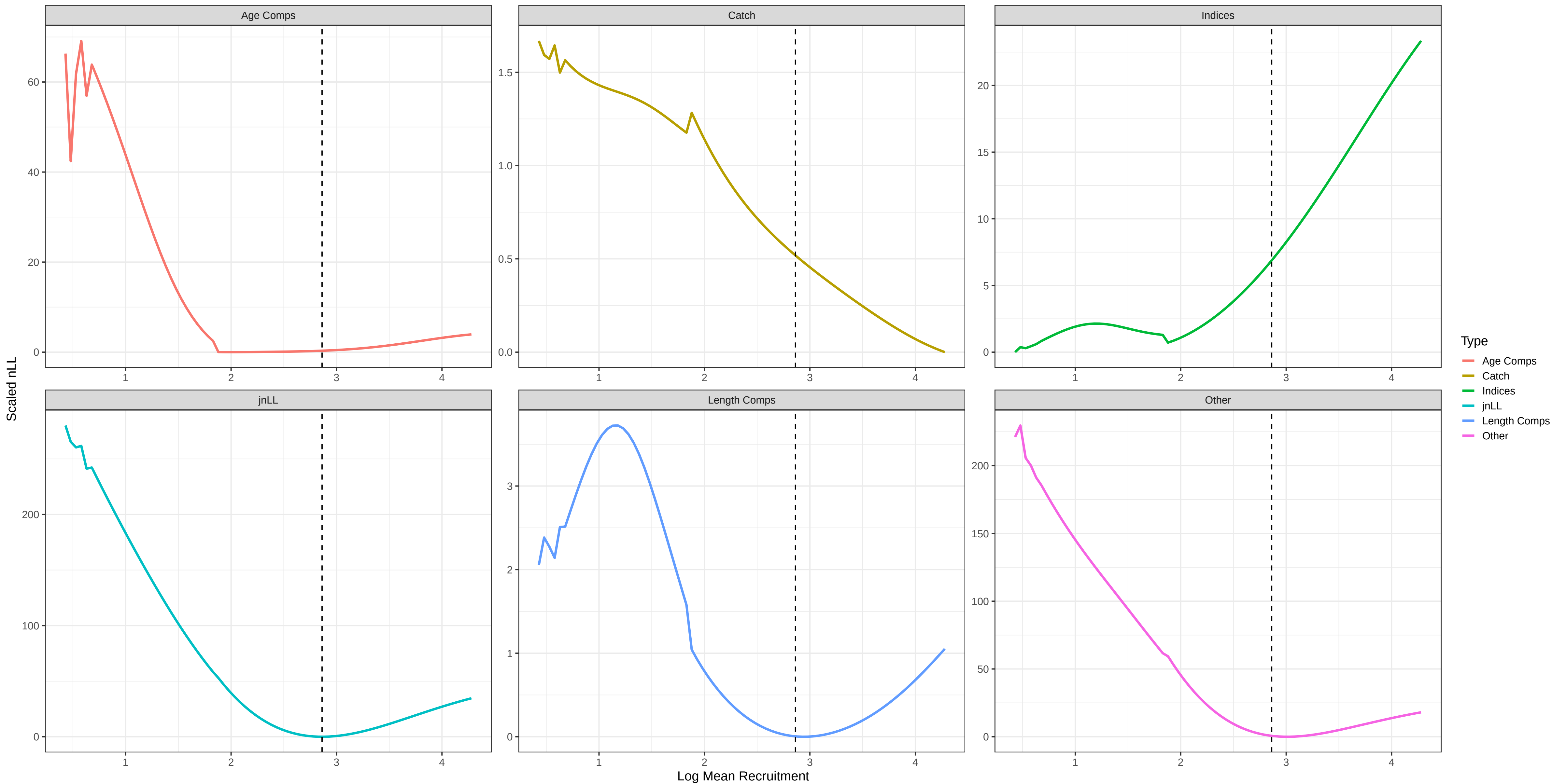




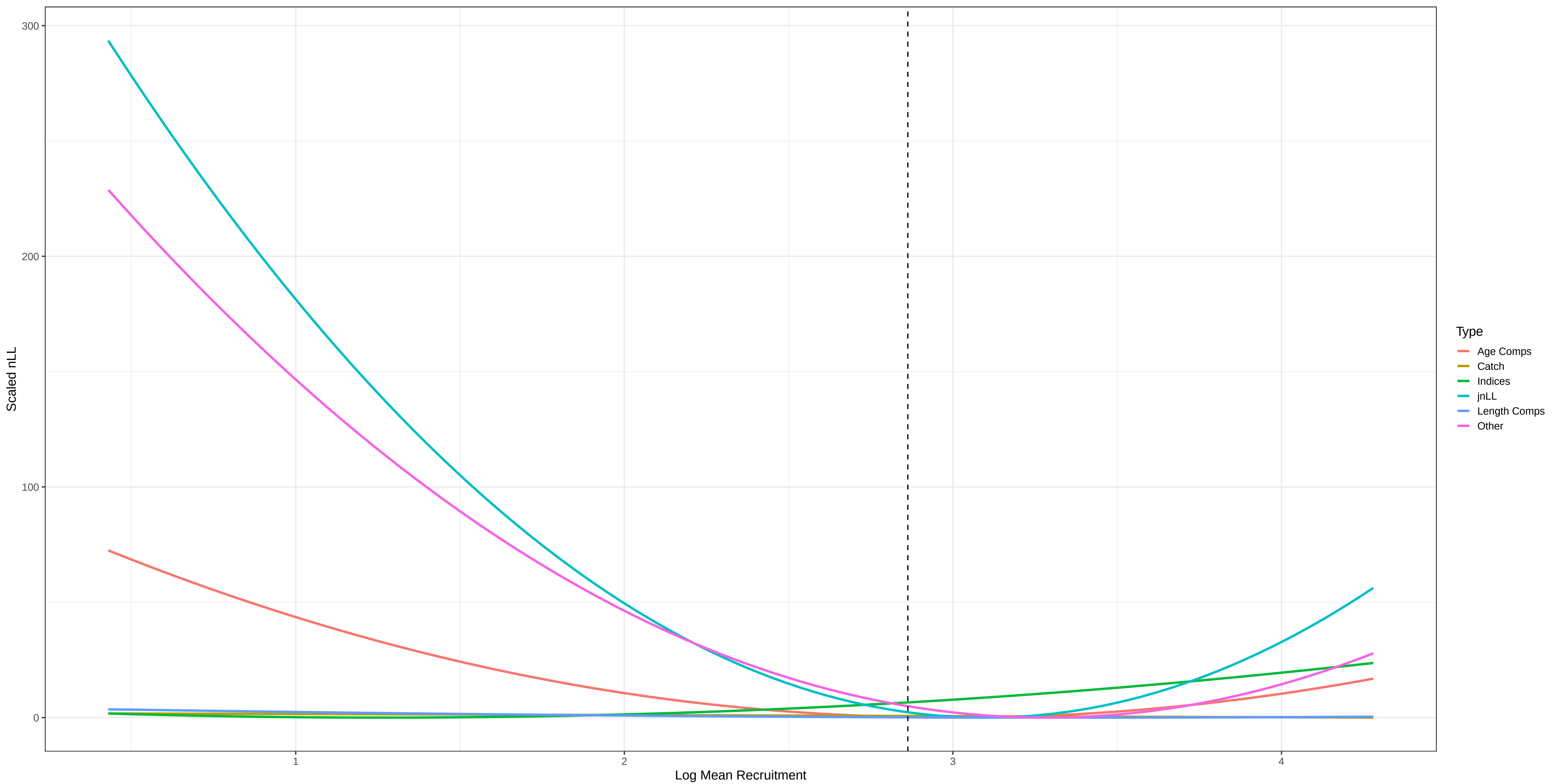
Profile Likelihood R0



Profile Likelihood R0

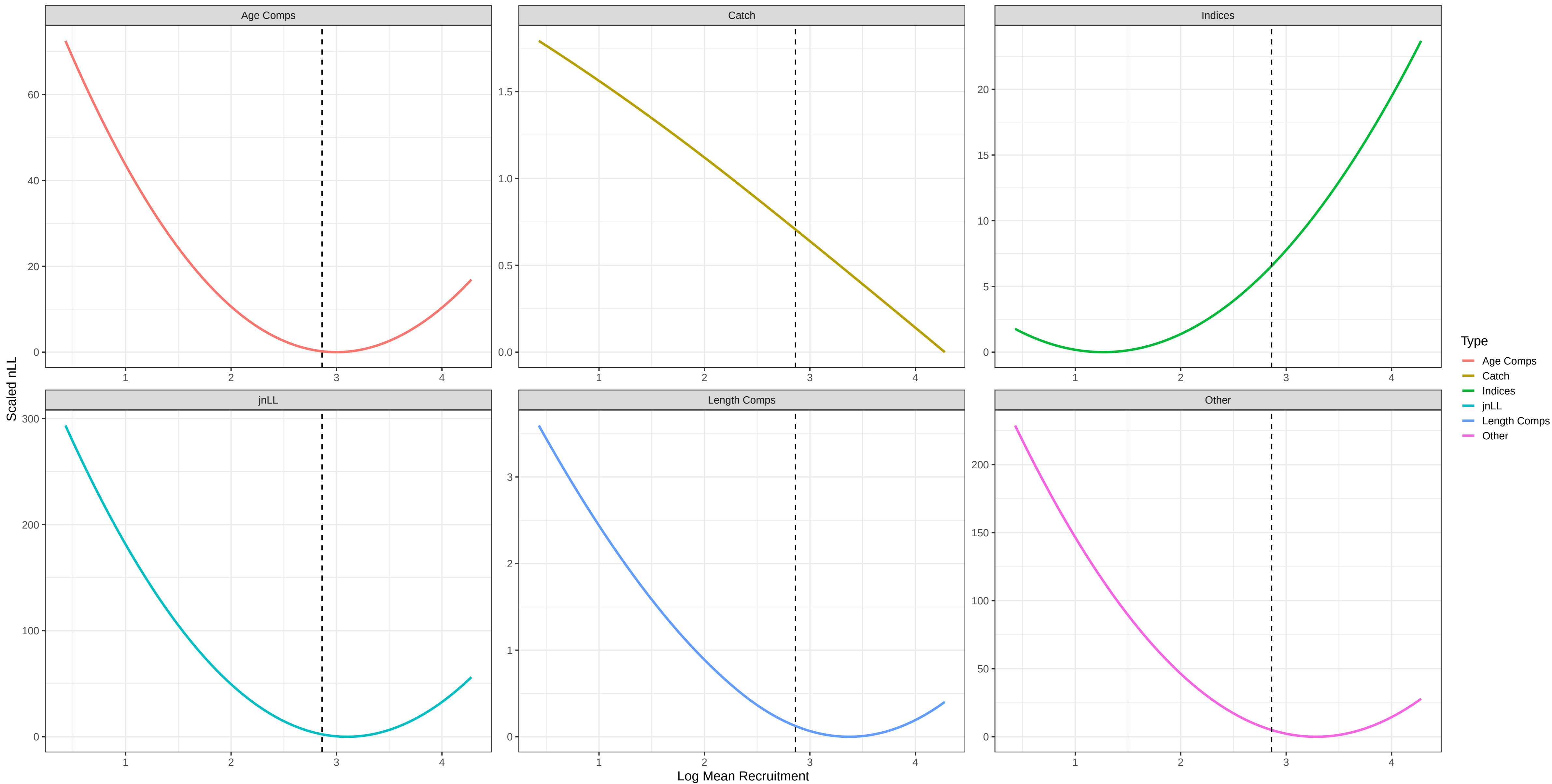


Profile Likelihood R0

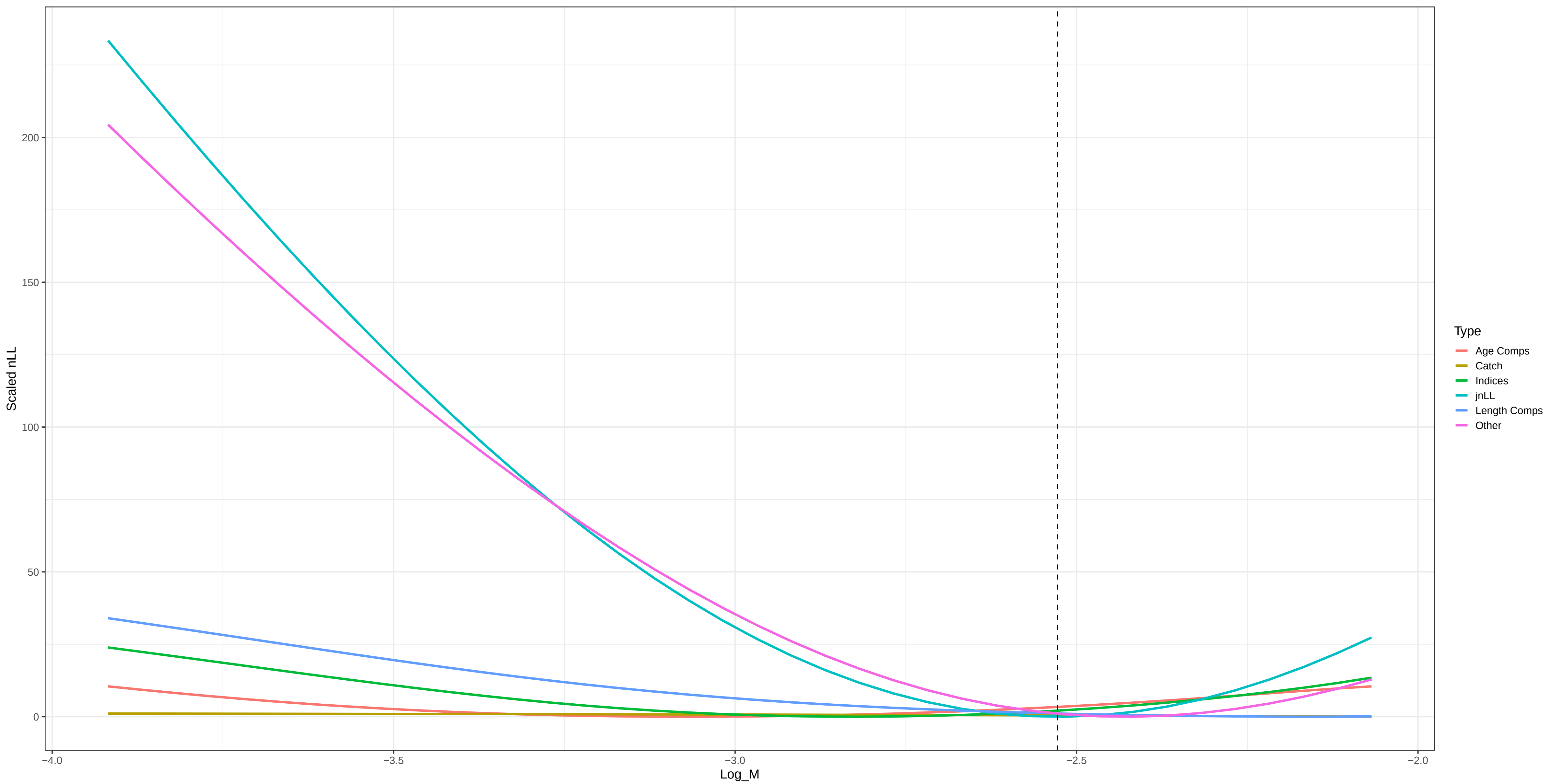




Profile Likelihood R0

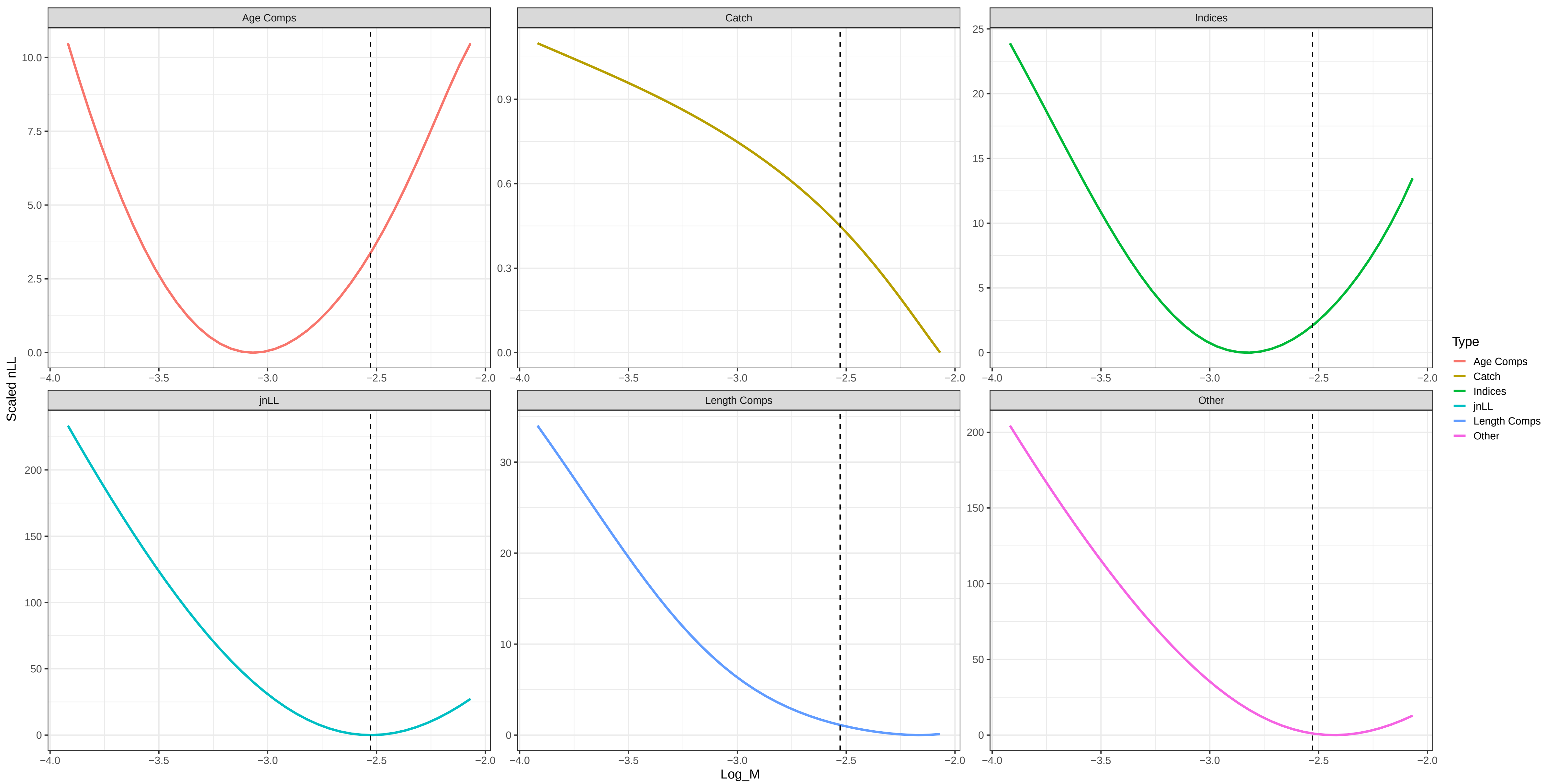


Profile Likelihood Female M

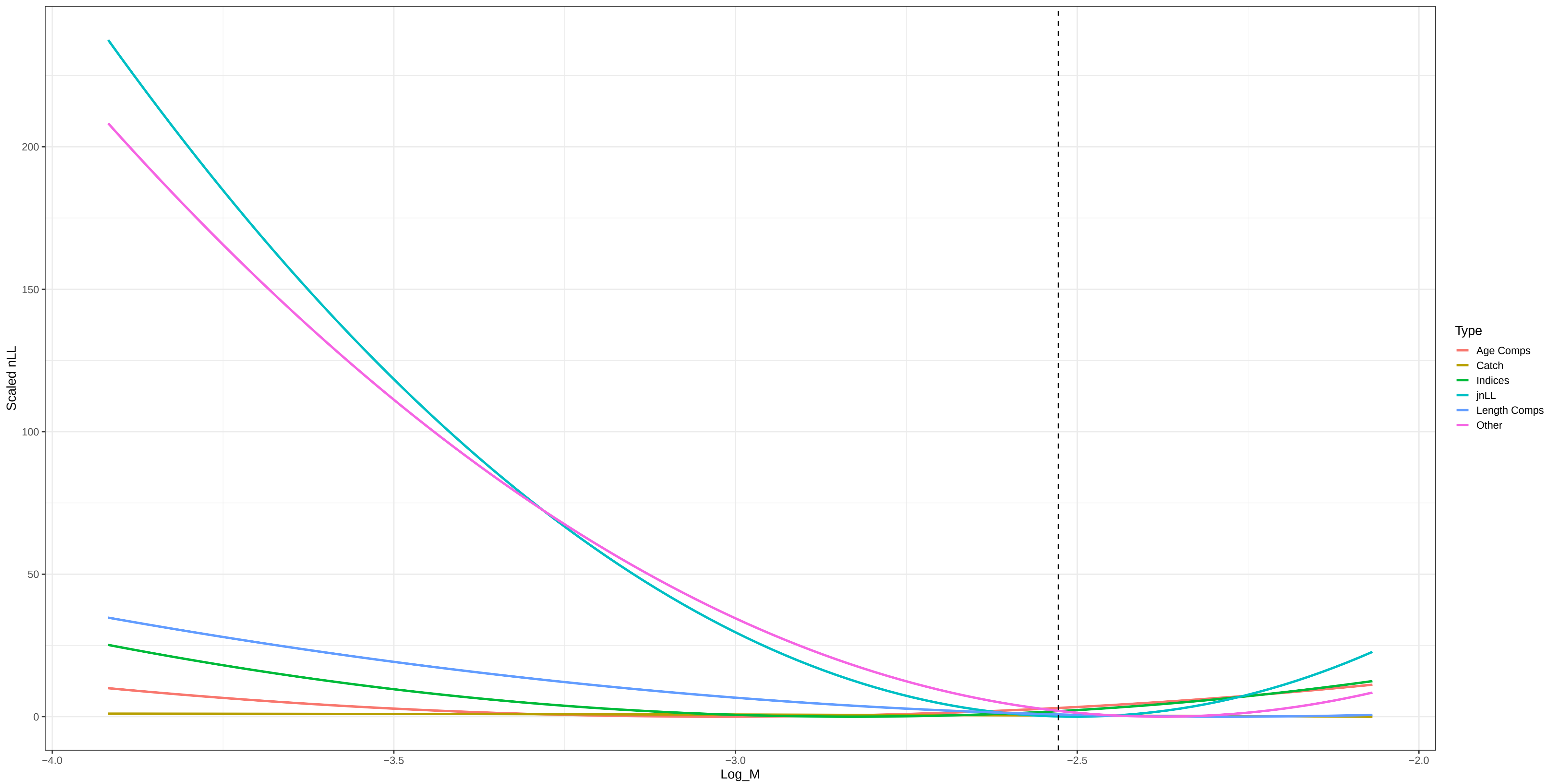




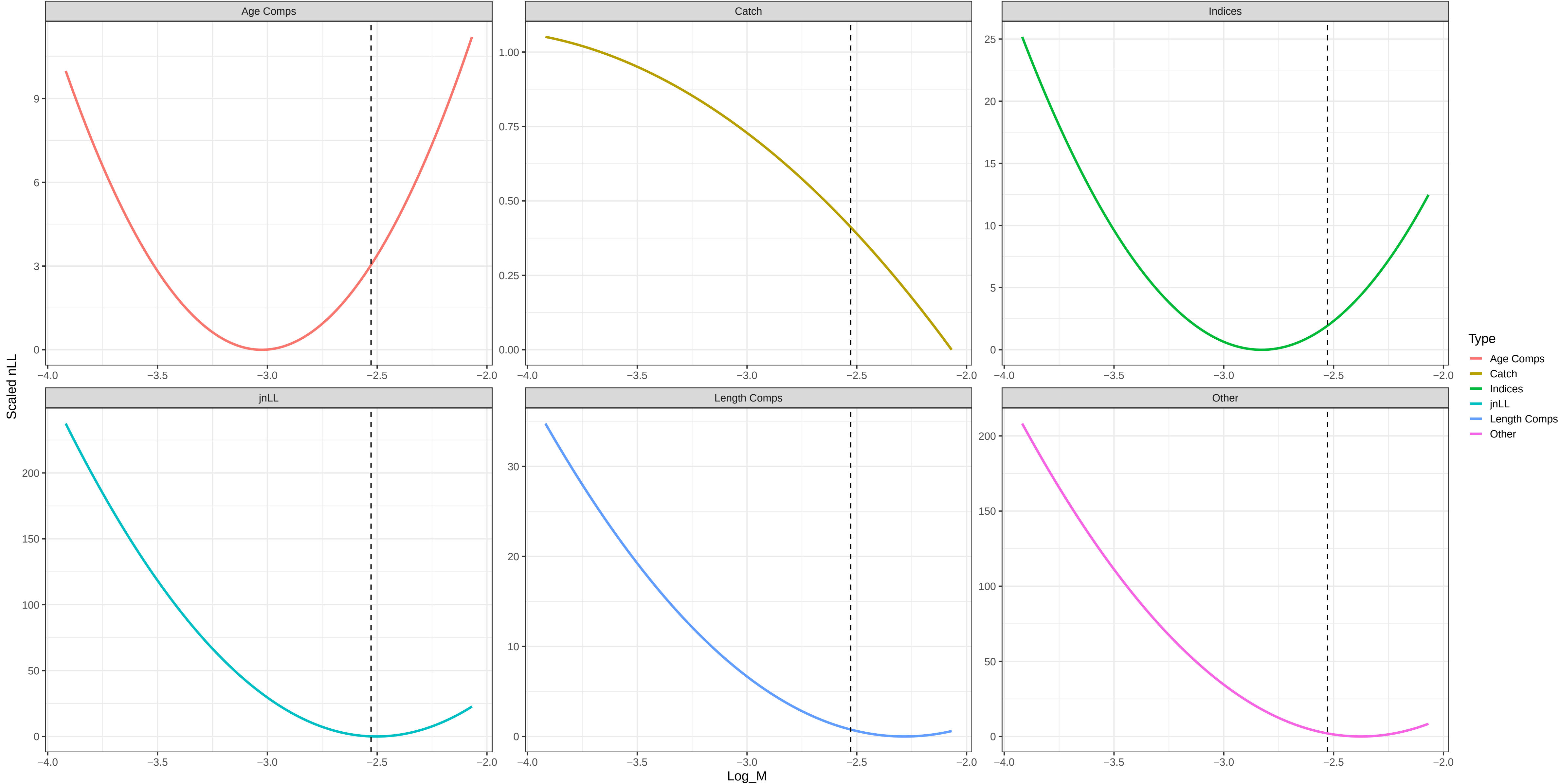
Profile Likelihood Female M



Profile Likelihood Female M

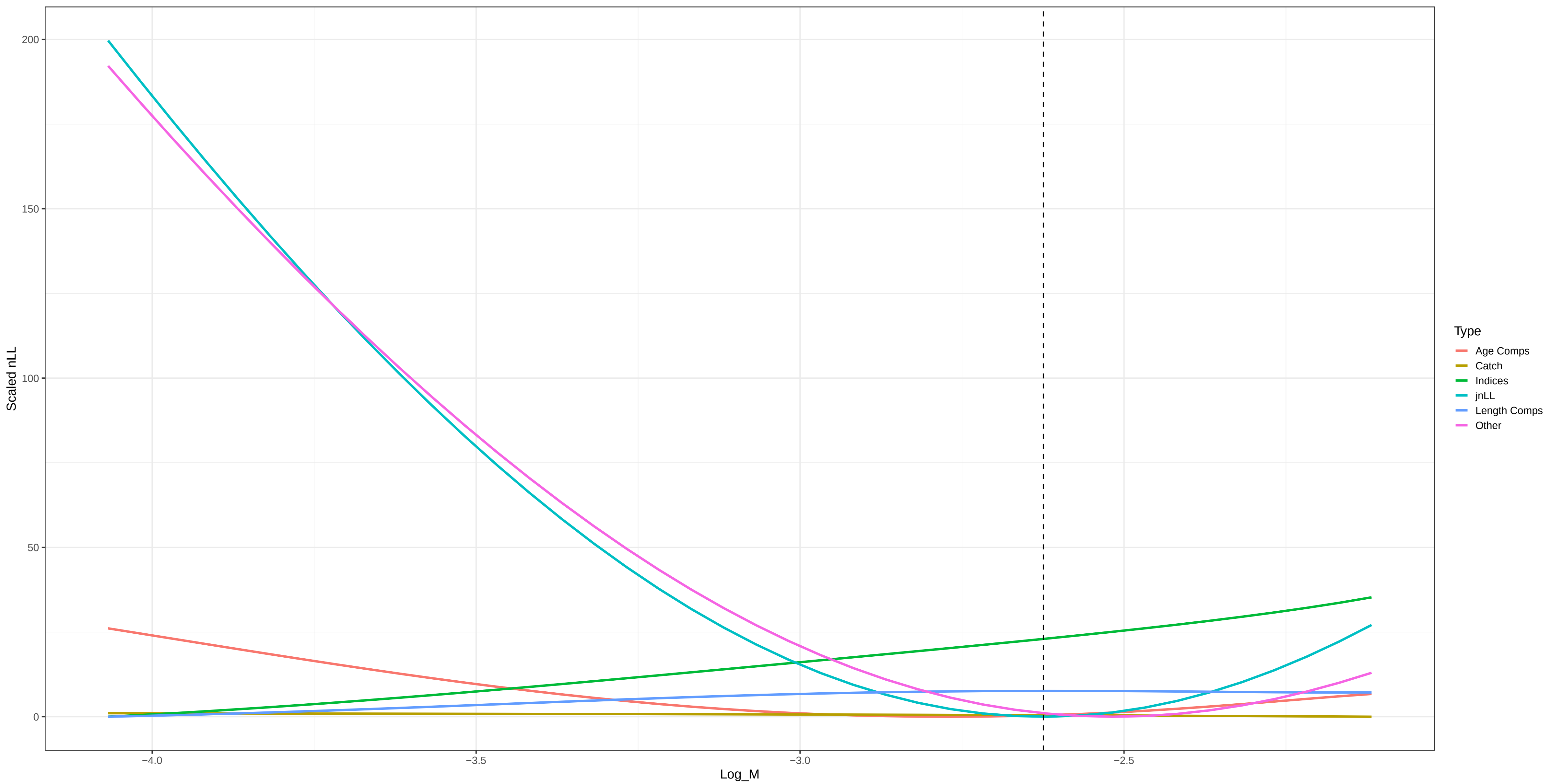


Profile Likelihood Female M

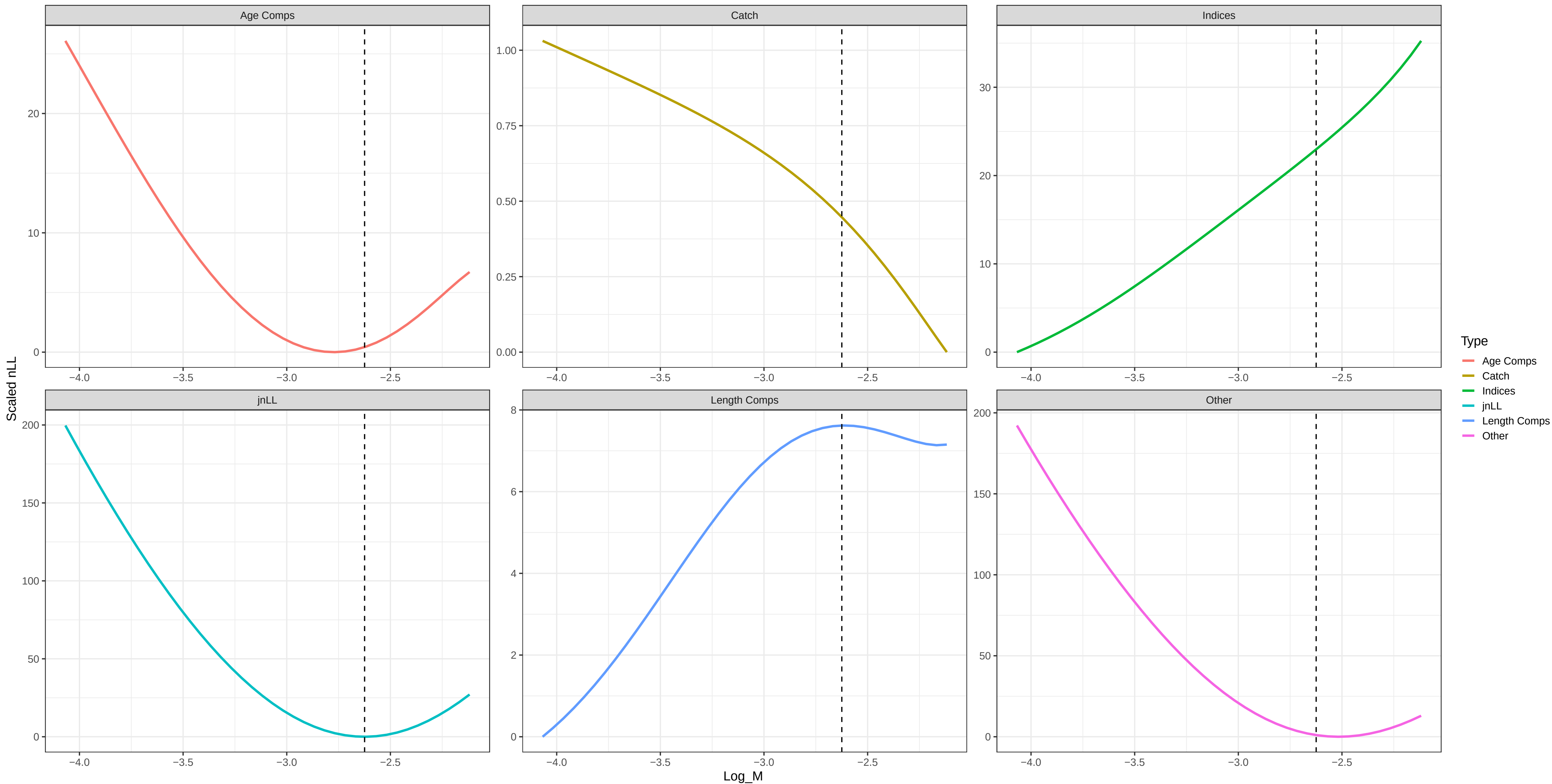




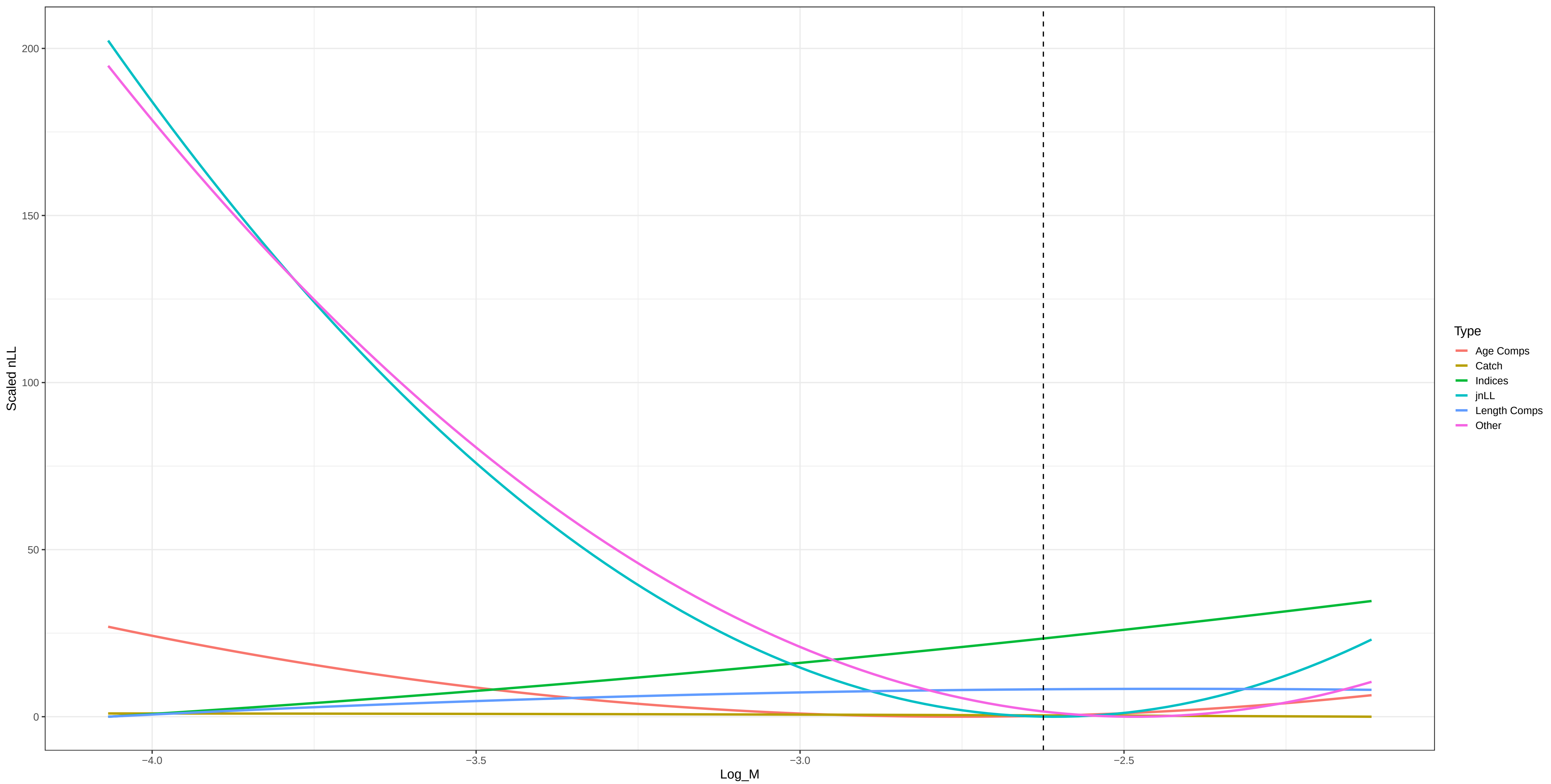
Profile Likelihood Male M



Profile Likelihood Male M



Profile Likelihood Male M





Profile Likelihood Male M

